

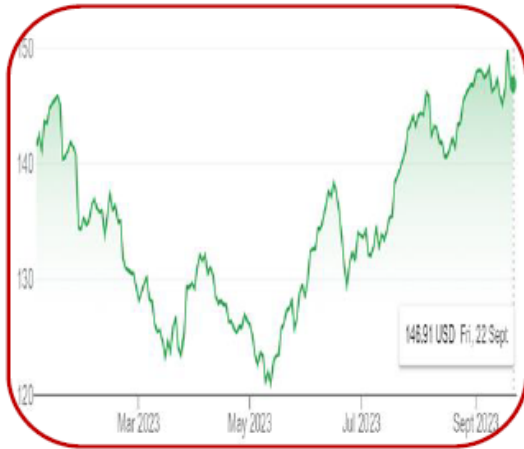
UNIT 1

Time Series Graphs

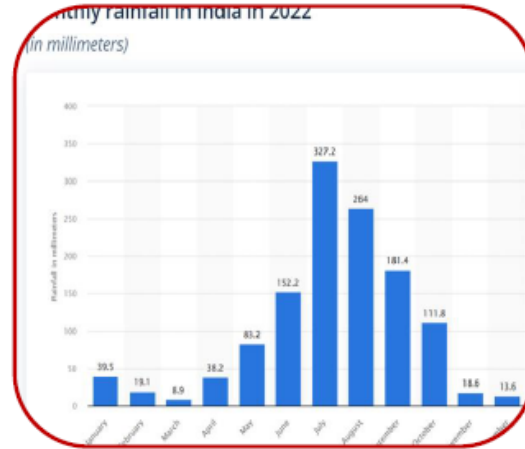
Hyndman, R.J., & Athanasopoulos, G. (2021) *Forecasting: principles and practice*, 3rd edition, OTexts: Melbourne, Australia. [OTexts.com/fpp3](https://otexts.com/fpp3). Accessed on

What is Time Series?

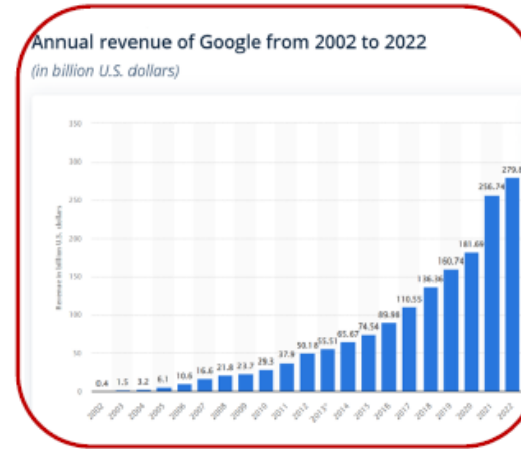
- A sequence of data points collected or recorded **at regular time intervals**, such as daily, monthly, or yearly.
- Time series analysis **helps uncover patterns, trends, and dependencies** within the data.



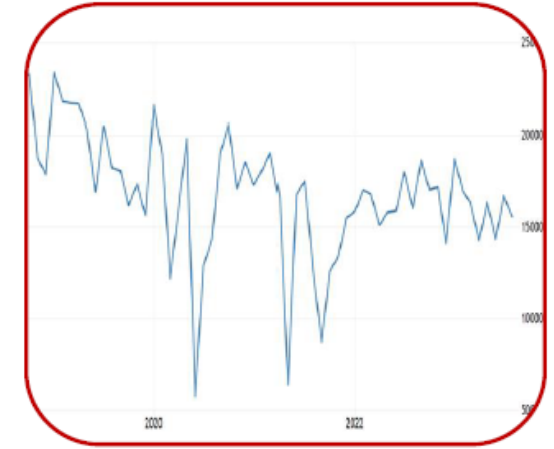
Daily IBM stock prices



Monthly rainfall



Annual Google Revenues



Car Production in India

Types of Data

Numerical (Quantitative) Data.



- Discrete or Continuous
- age, income, and temperature, sales, market price

Often involve statistical techniques

Categorical (Qualitative) Data.



- Represents categories or labels
- 2 types: nominal and ordinal
- gender, product types, and education levels

Requires frequency counts and chi-squared tests

Time Series Data.



- Data collected over time at regular intervals
- Temporal patterns
- stock prices, temperature records, and sales data

Time series regression and modeling methods

Text Data.



- Text or words
- unstructured and requires natural language processing (NLP) techniques for analysis
- articles, social media posts, and customer reviews

Requires NLP and text mining techniques

Data Collection Methods

Time Series

- collected at multiple time points over a continuous period
- gathered at regular intervals, such as daily, monthly, or yearly
- Common sources include sensors, financial markets, weather stations, and historical records

Cross-Sectional Data

- collected at a single point in time for multiple individuals, entities, or objects
- collected through surveys, questionnaires, or observations, where each observation represents a different unit or subject at the same point in time

Panel Data (Longitudinal Data)

- combination of both time series and cross-sectional data

Spatial Data

- Spatial data pertains to geographical locations and attributes associated with those locations
- Collected through GPS, Remote sensing, GIS

Forecasting data and methods

The appropriate forecasting methods depend largely on what data are available.

1. Qualitative forecasting

If there are no data available, or if the data available are not relevant to the forecasts, then **qualitative forecasting** methods must be used. These methods are not purely guesswork—there are well-developed structured approaches to obtaining good forecasts without using historical data.

2. Quantitative forecasting can be applied when two conditions are satisfied:

1. Numerical information about the past is available;
2. It is reasonable to assume that some aspects of the past patterns will continue into the future.

Time series forecasting

Examples of time series data include:

- Annual Google profits
- Quarterly sales results for Amazon
- Monthly rainfall
- Weekly retail sales
- Daily IBM stock prices
- Hourly electricity demand

Forecasting Example

When forecasting time series data, the aim is to estimate how the sequence of observations will continue into the future. Figure below shows the quarterly Australian beer production from 2000 to the second quarter of 2010.

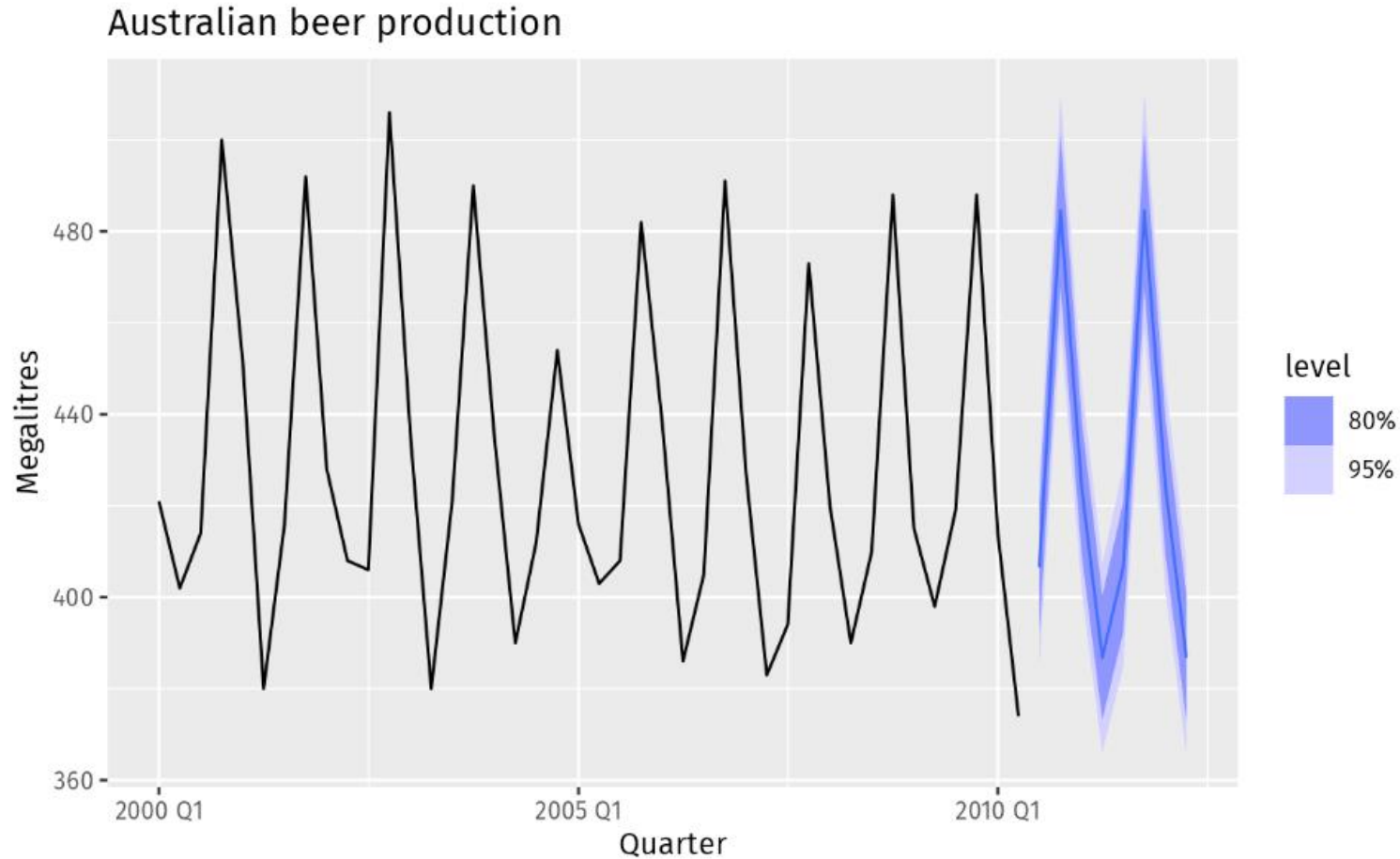


Figure 1: Australian quarterly beer production: 2000Q1–2010Q2, with two years of forecasts.

Predictor variables and time series forecasting

Explanatory model

$ED = f(\text{current temperature, strength of economy, population, time of day, day of week, error}).$

Time series model

$$ED_{t+1} = f(ED_t, ED_{t-1}, ED_{t-2}, ED_{t-3}, \dots, \text{error})$$

Mixed models

$$ED_{t+1} = f(ED_t, \text{current temperature, time of day, day of week, error})$$

The basic steps in a forecasting task

Step 1: Problem definition

Step 2: Gathering information

Step 3: Preliminary (exploratory) analysis

Step 4: Choosing and fitting models

Step 5: Using and evaluating a forecasting model

The basic steps in a forecasting task

**Understanding
Time Series**

**Time Series
Components –
Trend,
Seasonality,
Cyclicalilty,
Others**

**Time Series
Regression
Models (AR,
MA, ARIMA,
SARIMA,
SARIMAX)**

**Long Short
Term Memory
based models
(LSTM)**

Model Building

Model Validation

***VAR, Granger Causality,
Cointegration and
Multicollinearity, ARCH,
GARCH***

Time Series Graph

Plots allow us to identify:

- ▣ Patterns;
- ▣ Unusual observations;
- ▣ Changes over time;
- ▣ Relationships between variables.

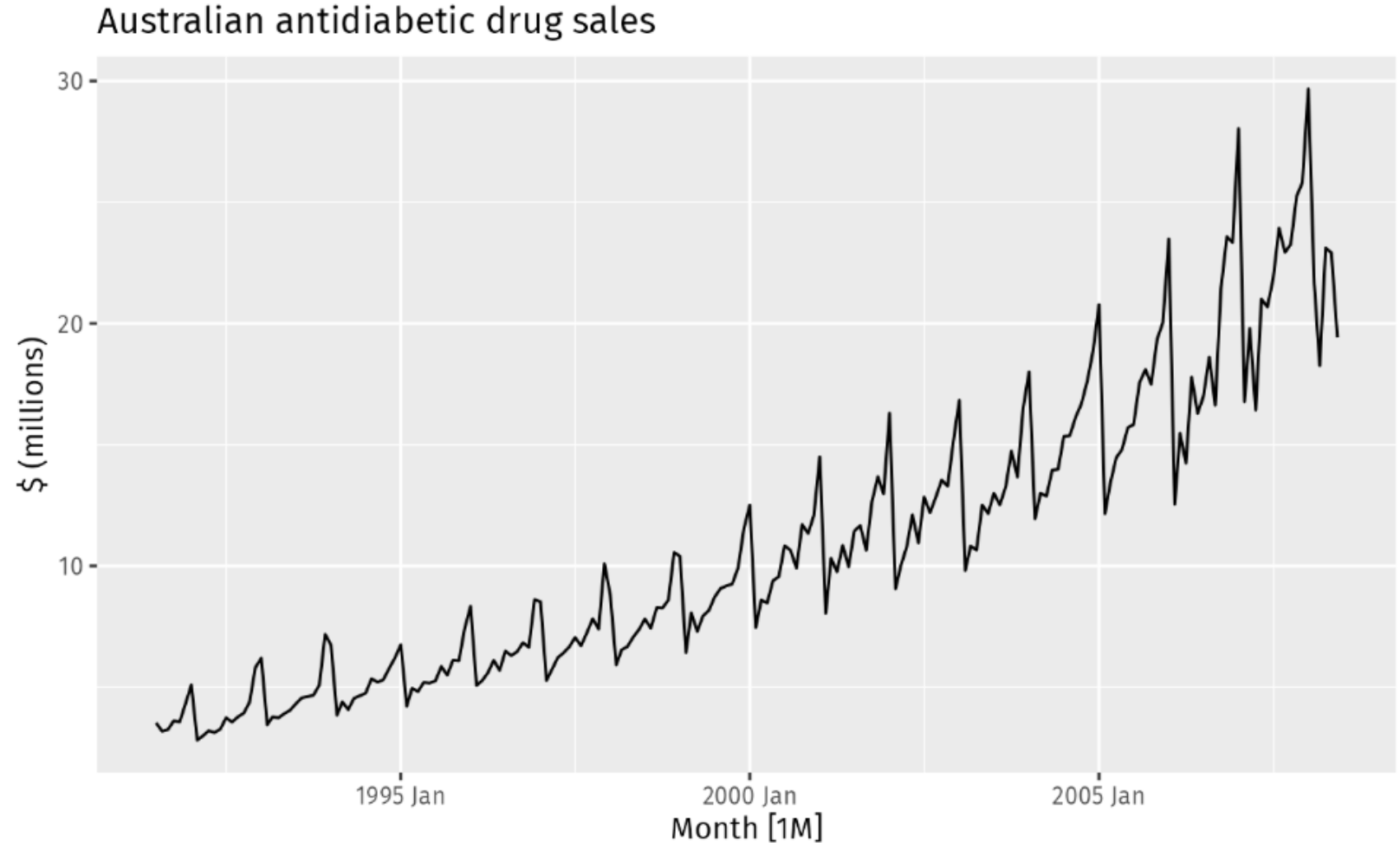
Australian antidiabetic drug sales

Here, there is a clear and increasing trend.

There is also a strong seasonal pattern that increases in size as the level of the series increases.

The sudden drop at the start of each year is caused by a government subsidisation scheme that makes it cost-effective for patients to stockpile drugs at the end of the calendar year.

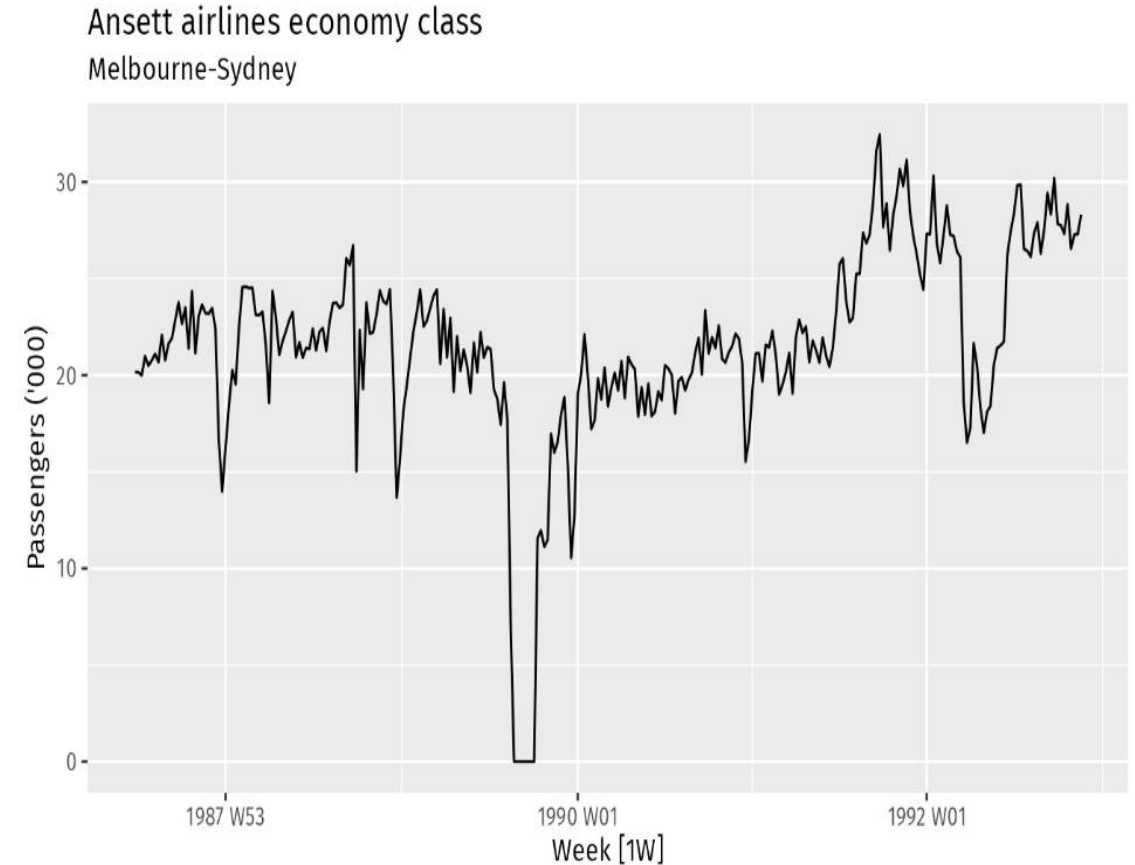
Any forecasts of this series would need to capture the seasonal pattern, and the fact that the trend is changing slowly.



Weekly economy passenger load on Ansett Airlines

The time plot immediately reveals some interesting features.

- There was a period in 1989 when no passengers were carried — this was due to an industrial dispute.
- There was a period of reduced load in 1992. This was due to a trial in which some economy class seats were replaced by business class seats.
- A large increase in passenger load occurred in the second half of 1991.
- There are some large dips in load around the start of each year. These are due to holiday effects.
- There is a long-term fluctuation in the level of the series which increases during 1987, decreases in 1989, and increases again through 1990 and 1991.



Time series patterns

Components of a Time Series

- Introduce the components that make up a time series:

Trend: The long-term movement or underlying pattern.

Seasonality:
Repeating patterns at regular intervals.

Cyclical Variations:
Longer-term periodic fluctuations.

Irregular/Residual
: Random noise or irregularities not explained by other components.

- These components combine to form the observed time series.

Trend pattern exists when there is a long-term increase or decrease in the data.

Seasonal pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Cyclic pattern exists when data exhibit rises and falls that are *not of fixed period* (duration usually of at least 2 years).

Seasonal Vs Cyclic

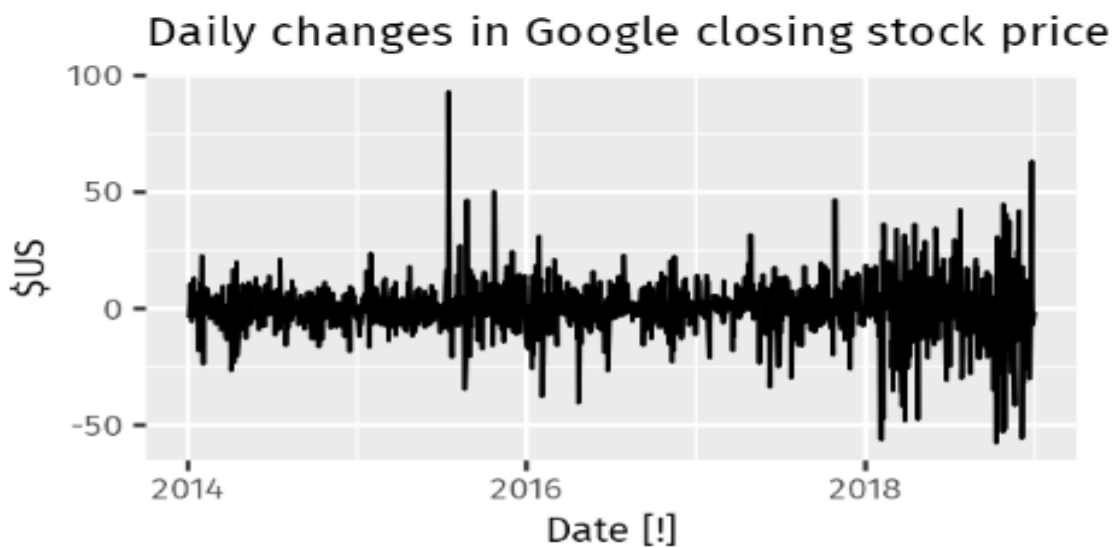
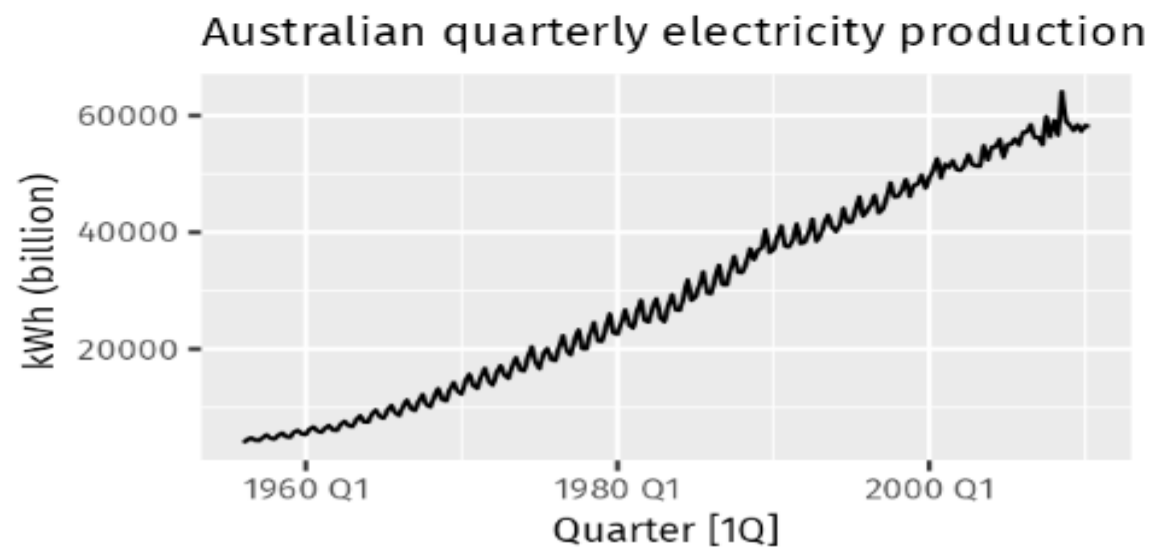
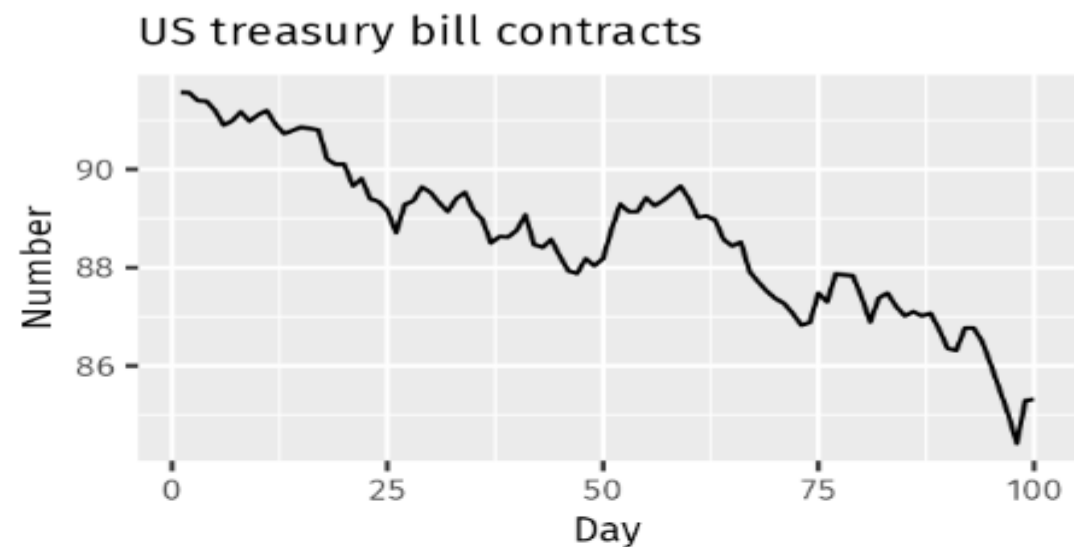
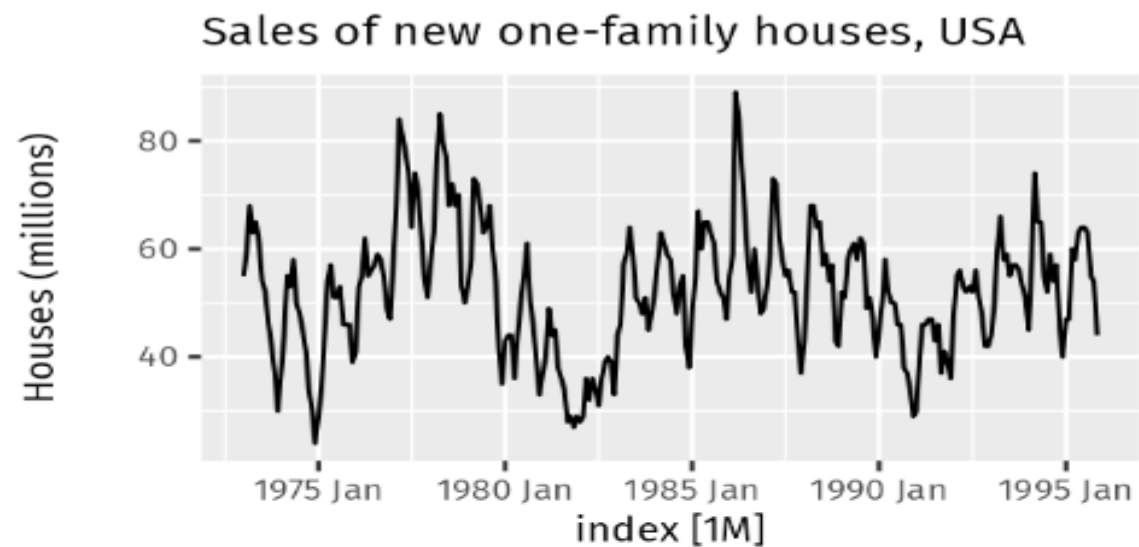
Many people confuse cyclic behavior with seasonal behavior, but they are really quite different.

If the fluctuations are not of a fixed frequency then they are cyclic; if the frequency is unchanging and associated with some aspect of the calendar, then the pattern is seasonal.

In general, the average length of cycles is longer than the length of a seasonal pattern, and the magnitudes of cycles tend to be more variable than the magnitudes of seasonal patterns.

The timing of peaks and troughs is predictable with seasonal data, but unpredictable in the long term with cyclic data.

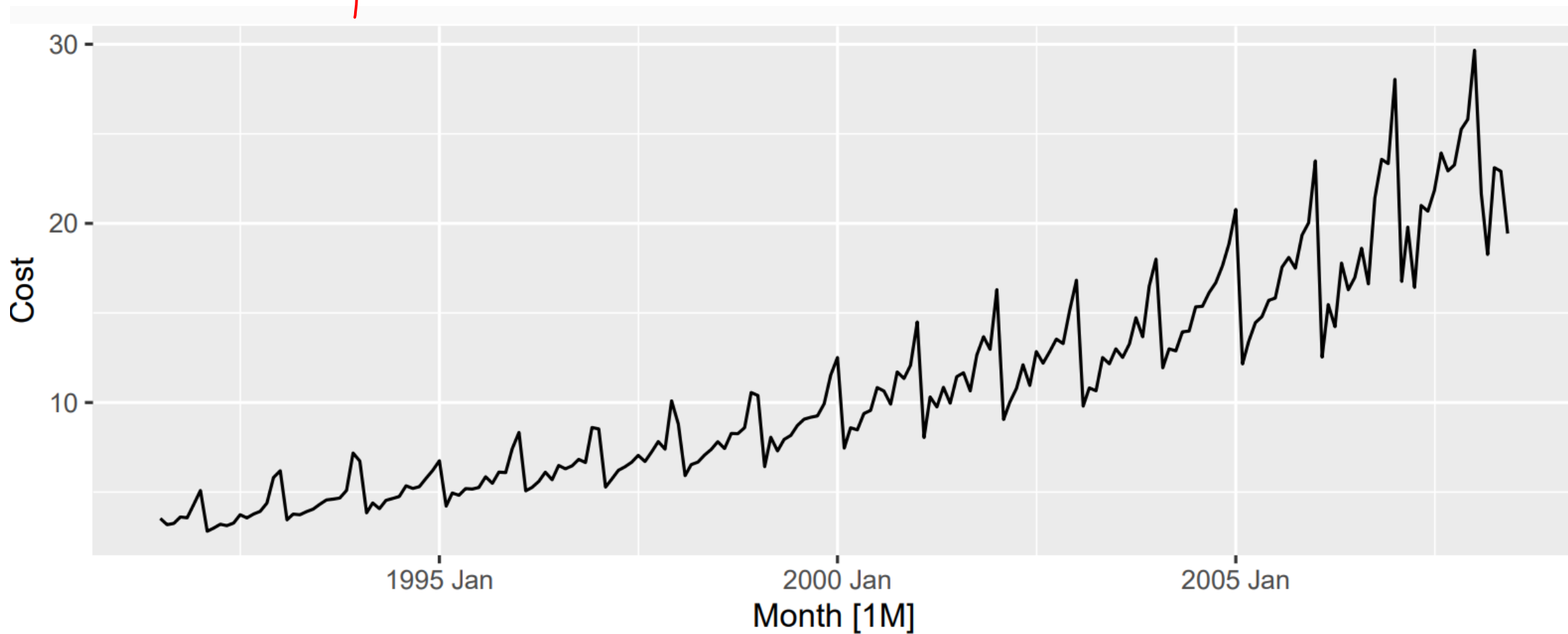
The examples in Figure show different combinations of these components.



- 1.The monthly housing sales (top left) show strong seasonality within each year, as well as some strong cyclic behaviour with a period of about 6–10 years. There is no apparent trend in the data over this period.
- 2.The US treasury bill contracts (top right) show results from the Chicago market for 100 consecutive trading days in 1981. Here there is no seasonality, but an obvious downward trend. Possibly, if we had a much longer series, we would see that this downward trend is actually part of a long cycle, but when viewed over only 100 days it appears to be a trend.
- 3.The Australian quarterly electricity production (bottom left) shows a strong increasing trend, with strong seasonality. There is no evidence of any cyclic behaviour here.
- 4.The daily change in the Google closing stock price (bottom right) has no trend, seasonality or cyclic behaviour. There are random fluctuations which do not appear to be very predictable, and no strong patterns that would help with developing a forecasting model.

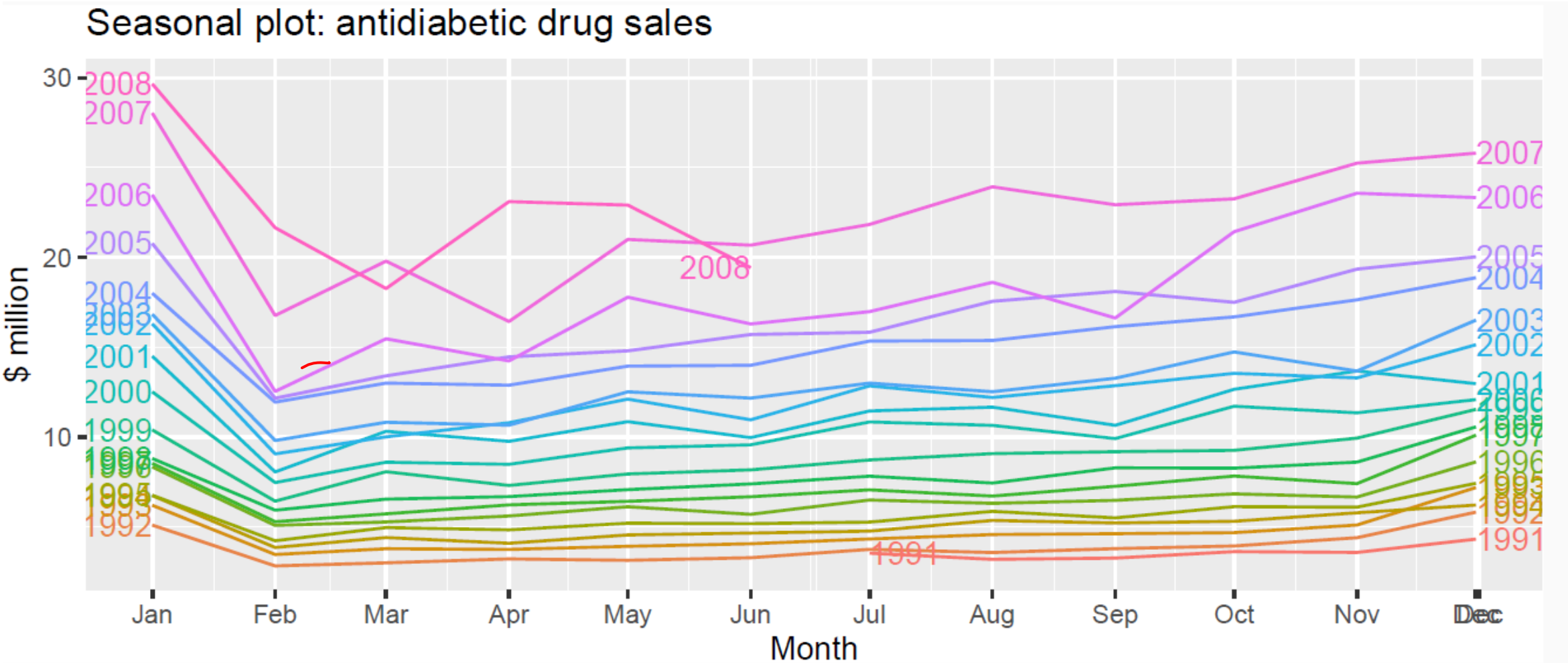
Seasonal plots

Time plot



Seasonal plots

A seasonal plot is similar to a time plot except that the data are plotted against the individual “seasons” in which the data were observed. An example is given in Figure showing the antidiabetic drug sales.



These are exactly the same data as were shown earlier, but now the data from each season are overlapped.

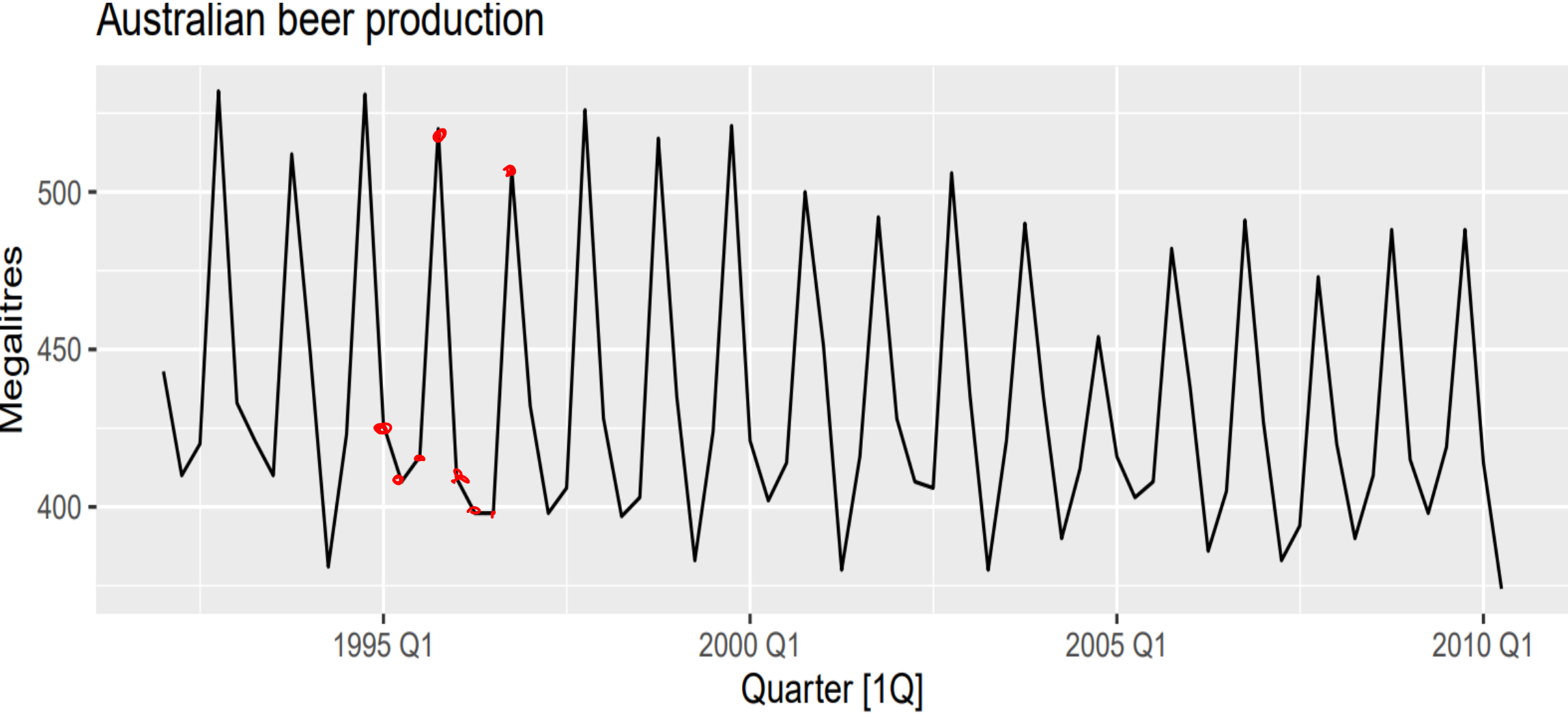
A seasonal plot allows the underlying seasonal pattern to be seen more clearly, and is especially useful in identifying years in which the pattern changes.

In this case, it is clear that there is a large jump in sales in January each year.

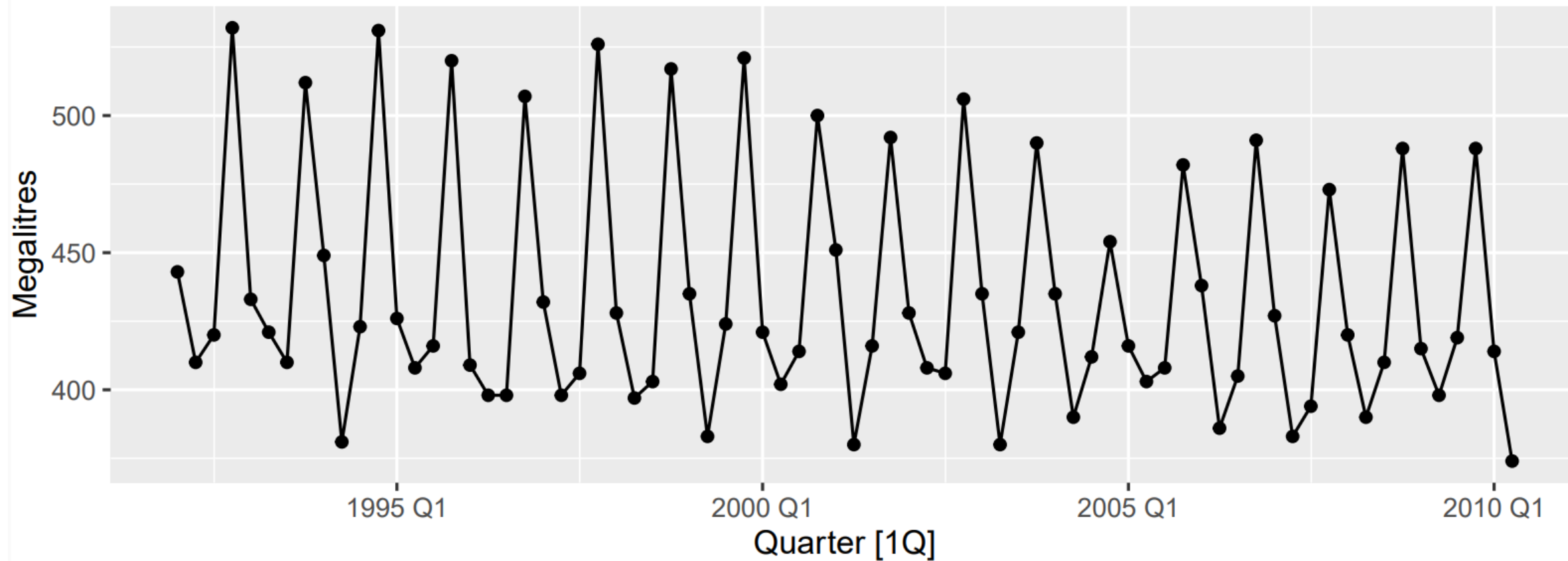
Actually, these are probably sales in late December as customers stockpile before the end of the calendar year, but the sales are not registered with the government until a week or two later.

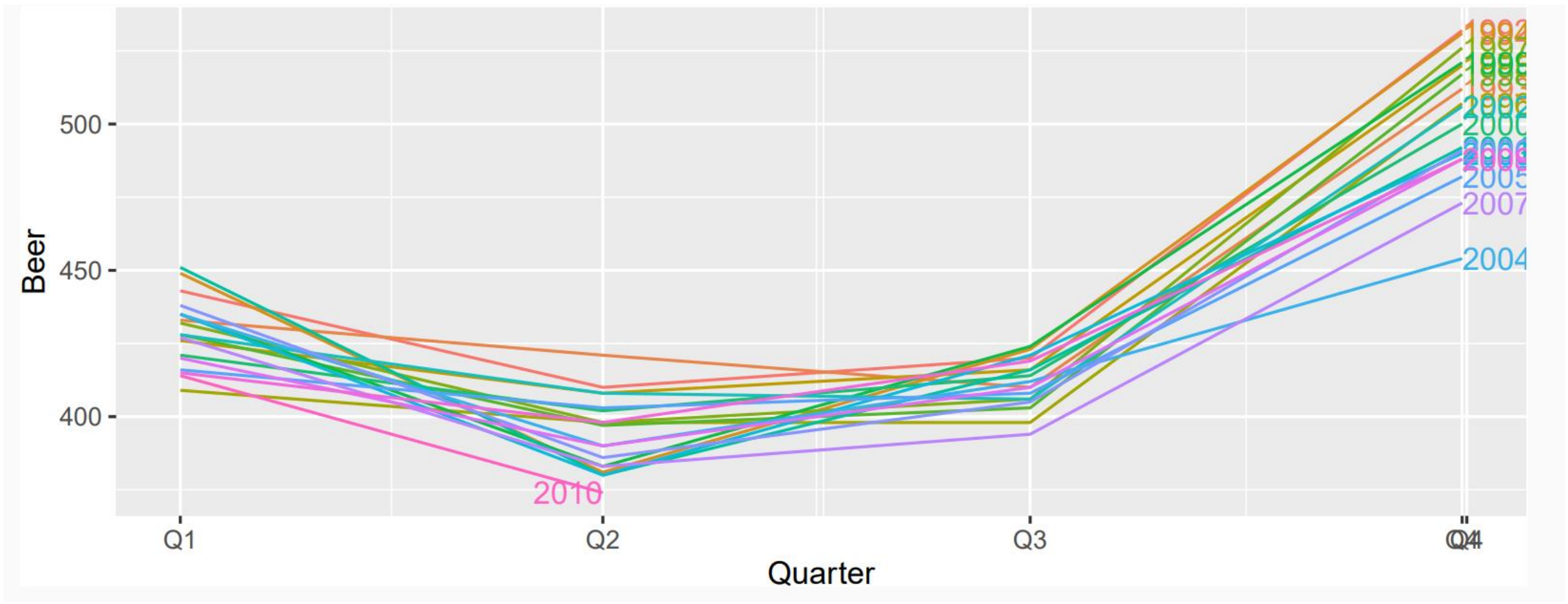
The graph also shows that there was an unusually small number of sales in March 2008 (most other years show an increase between February and March).

The small number of sales in June 2008 is probably due to incomplete counting of sales at the time the data were collected.



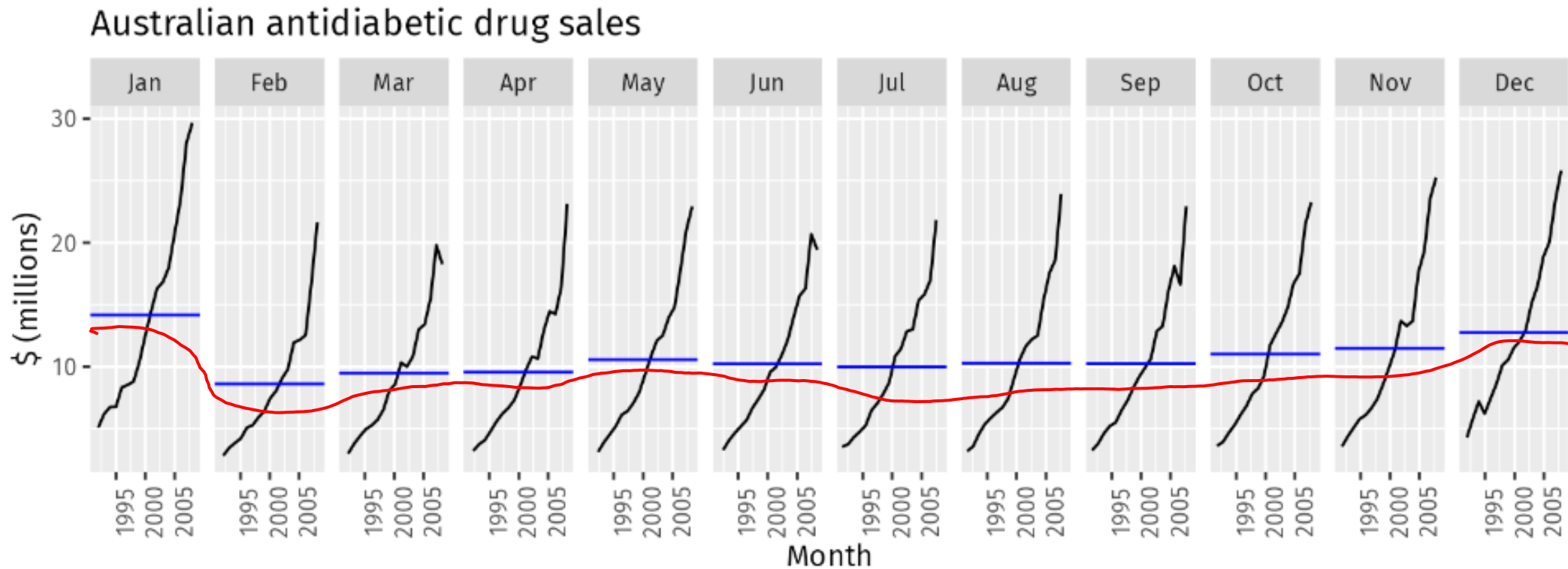
Australian beer production





Seasonal subseries plots

An alternative plot that emphasises the seasonal patterns is where the data for each season are collected together in separate mini time plots.



The blue horizontal lines indicate the means for each month. This form of plot enables the underlying seasonal pattern to be seen clearly, and also shows the changes in seasonality over time. It is especially useful in identifying changes within particular seasons. In this example, the plot is not particularly revealing; but in some cases, this is the most useful way of viewing seasonal changes over time.

Example: Australian holiday tourism

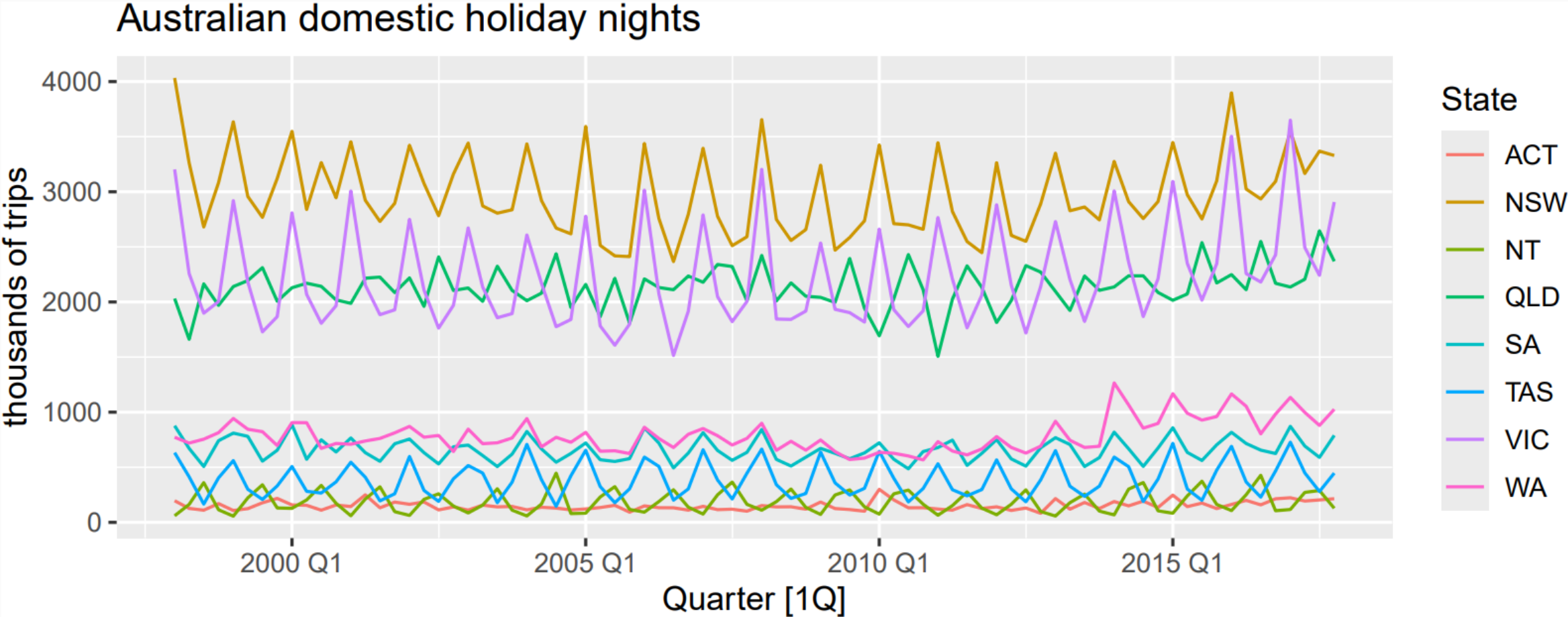
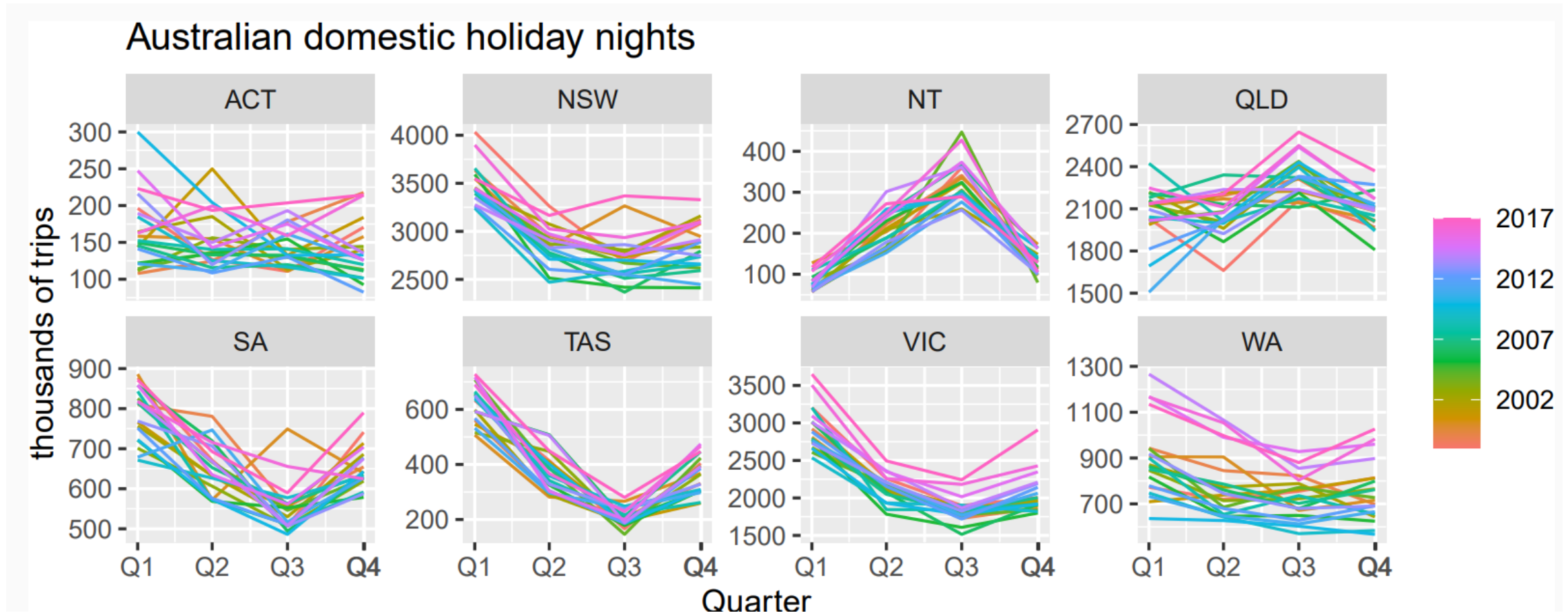


Figure: Time plots of Australian domestic holidays by state

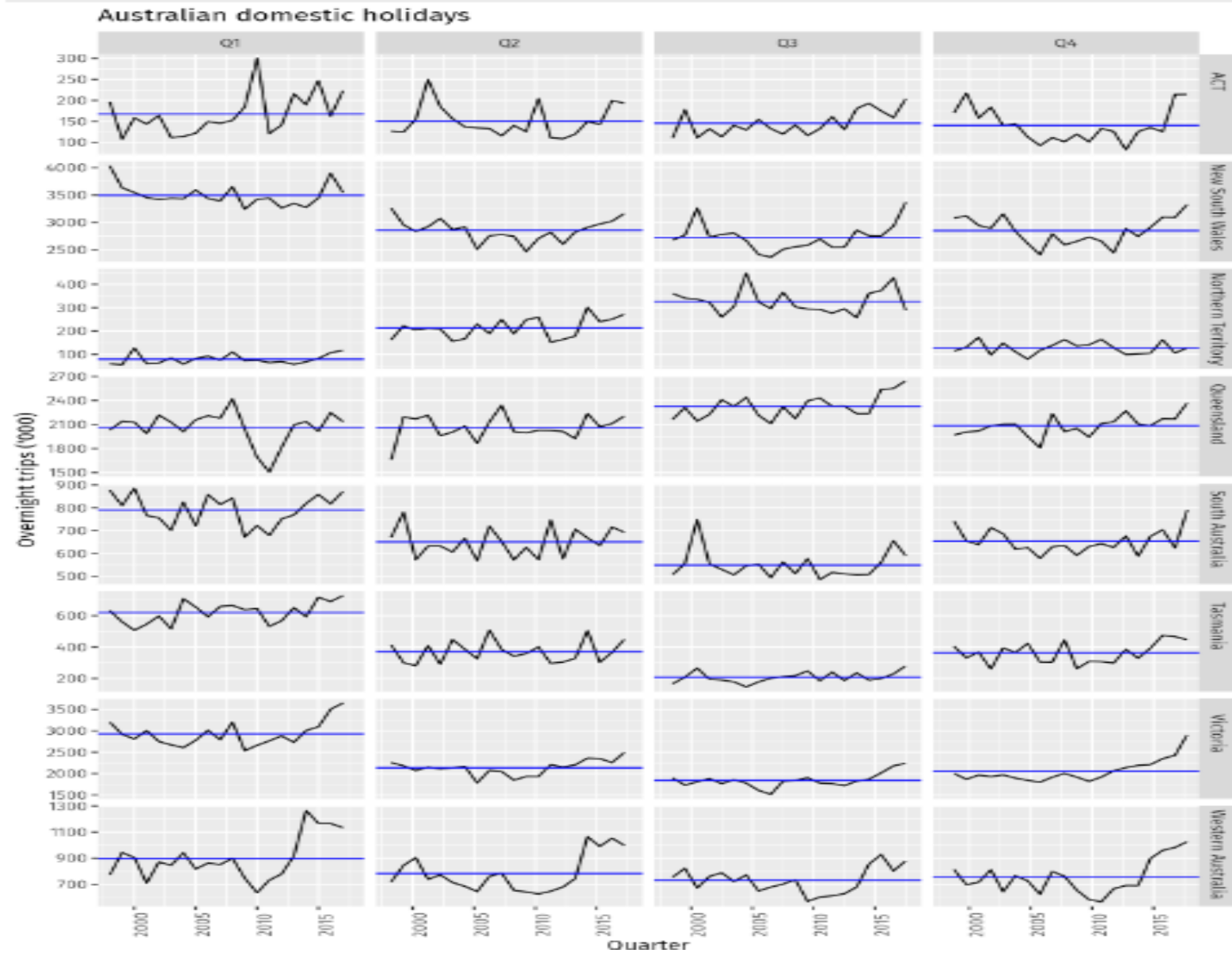
Seasonal plots

To see the timing of the seasonal peaks in each state, we can use a season plot.

Figure [2.10](#) makes it clear that the southern states of Australia (Tasmania, Victoria and South Australia) have strongest tourism in Q1 (their summer), while the northern states (Queensland and the Northern Territory) have the strongest tourism in Q3 (their dry season).



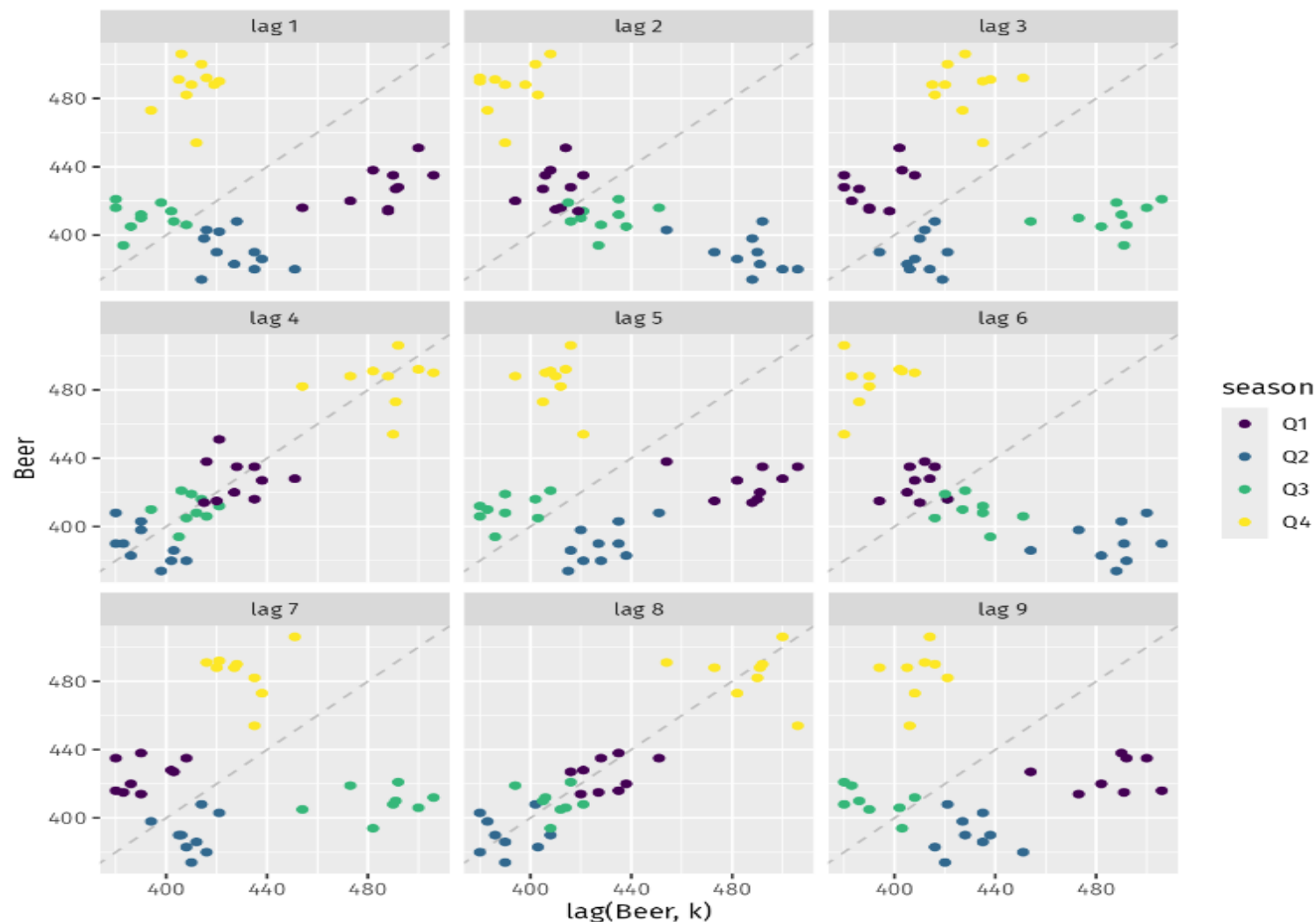
Seasonal subseries plots



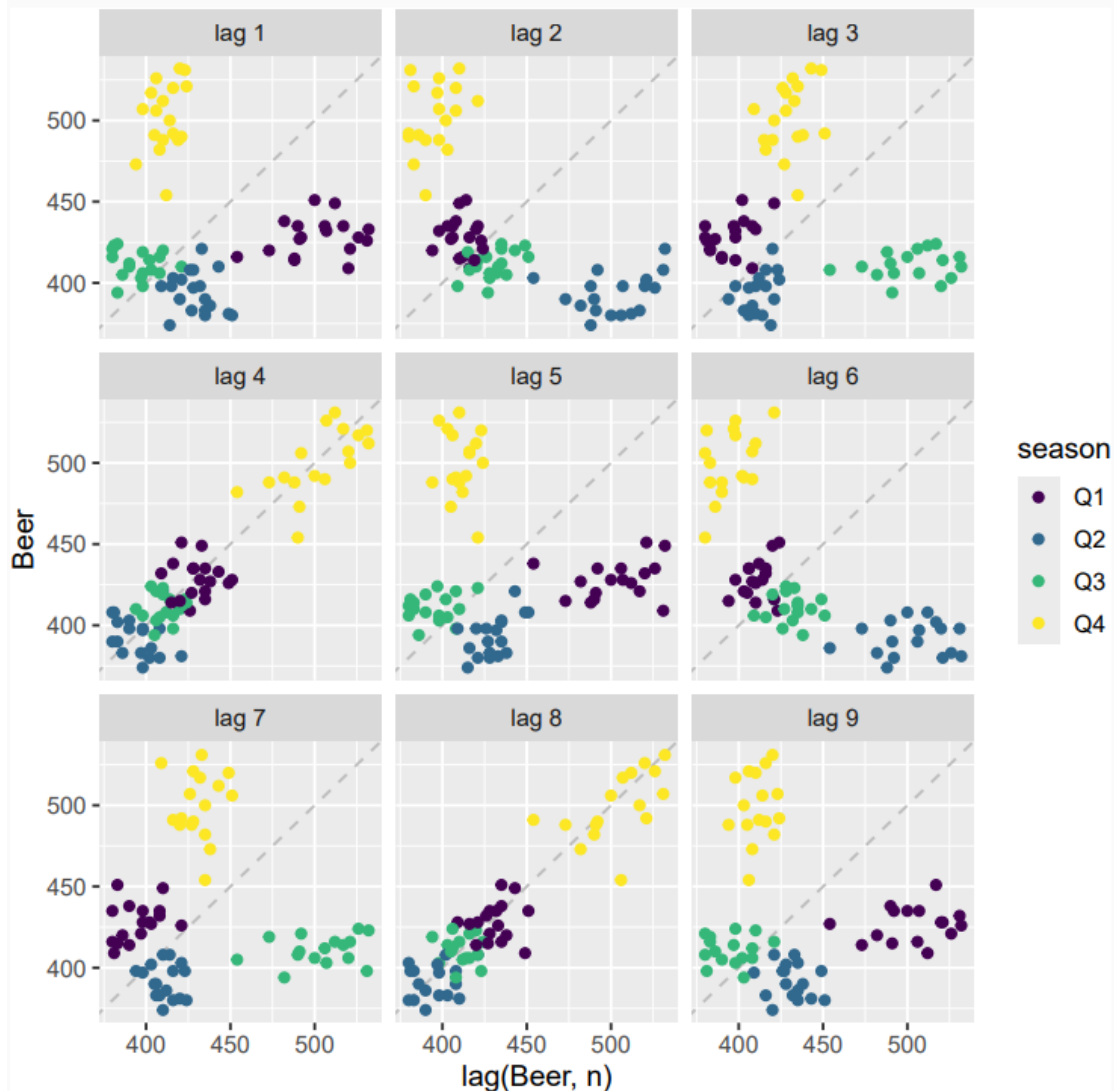
Lag plots

Figure displays scatterplots of quarterly Australian beer production (introduced in earlier slides), where the horizontal axis shows lagged values of the time series. Each graph shows y_t plotted against y_{t-k} for different values of k .

Here the colours indicate the quarter of the variable on the vertical axis. The relationship is strongly positive at lags 4 and 8, reflecting the strong seasonality in the data. The negative relationship seen for lags 2 and 6 occurs because peaks (in Q4) are plotted against troughs (in Q2)



Autocorrelation



- Each graph shows y_t plotted against y_{t-k} for different values of k .
- The autocorrelations are the correlations associated with these scatterplots.
- $r_1 = \text{Correlation}(y_t, y_{t-1})$
- $r_2 = \text{Correlation}(y_t, y_{t-2})$
- $r_3 = \text{Correlation}(y_t, y_{t-3})$
- $\vdots$

Just as correlation measures the extent of a linear relationship between two variables, autocorrelation measures the linear relationship between *lagged values* of a time series.

There are several autocorrelation coefficients, corresponding to each panel in the lag plot. For example, r_1 measures the relationship between y_t and y_{t-1} , r_2 measures the relationship between y_t and y_{t-2} , and so on.

The value of r_k can be written as

$$r_k = \frac{\sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2} = \frac{\text{Cov}(y_t, y_{t-k})}{\text{VAR}(y_t)}$$

We denote the sample autocovariance at lag k by c_k and the sample autocorrelation at lag k by r_k . Then define

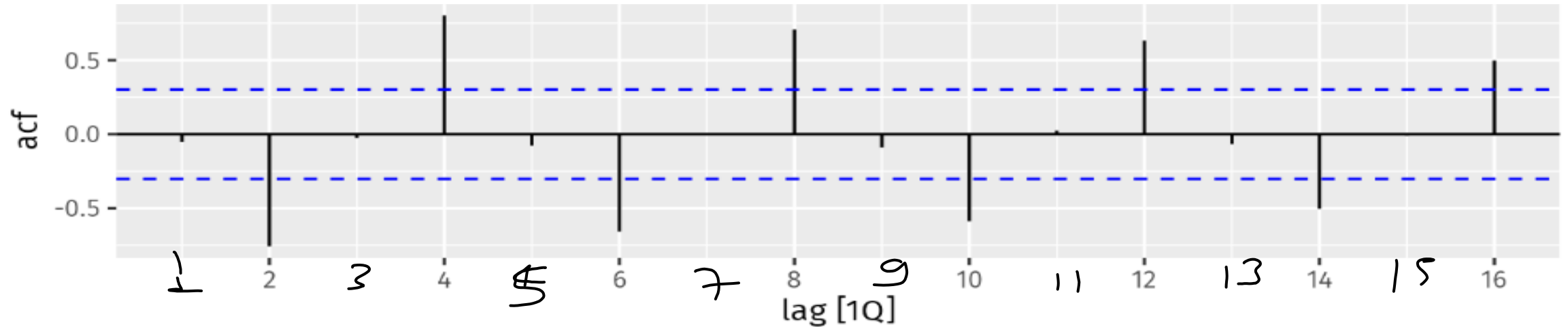
$$c_k = \frac{1}{T} \sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})$$

and

$$r_k = c_k / c_0$$

Autocorrelation plot (ACF Plot)

Together, the autocorrelations at lags 1, 2, $\dots$, make up the autocorrelation or ACF.
The plot is known as a correlogram

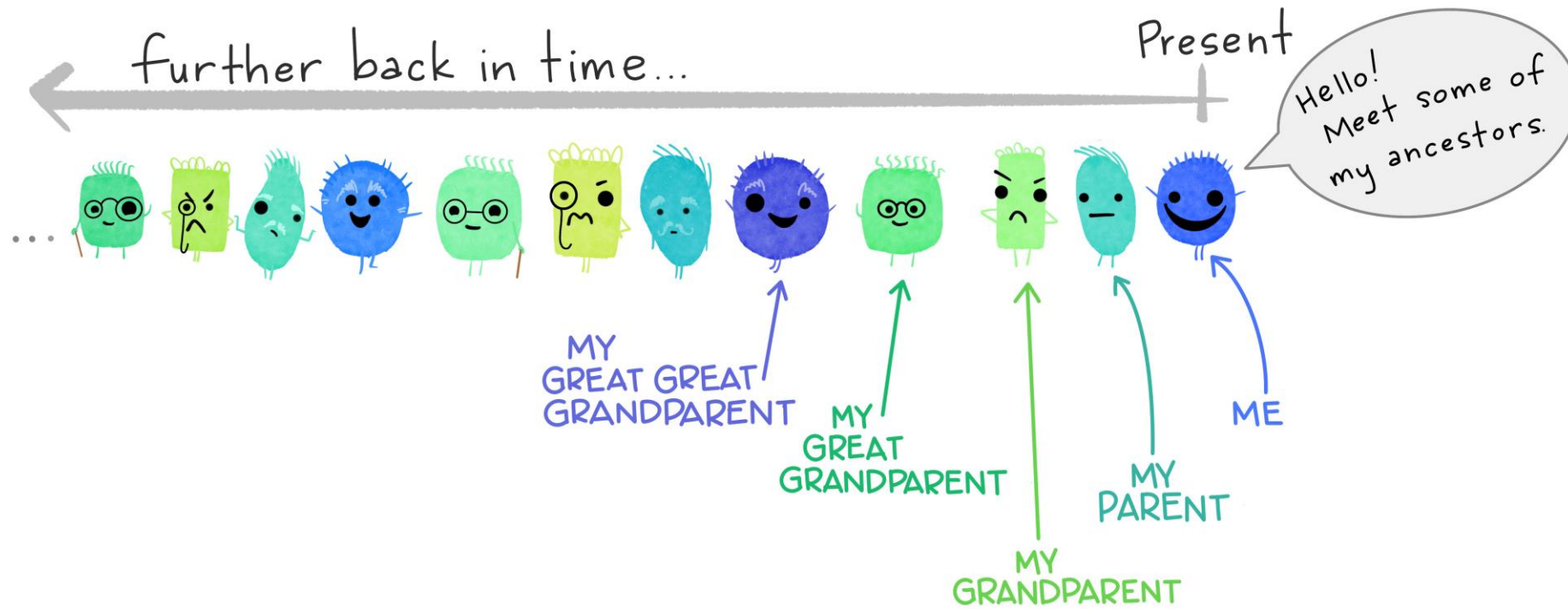


In this graph:

- r_4 is higher than for the other lags. This is due to the seasonal pattern in the data: the peaks tend to be four quarters apart and the troughs tend to be four quarters apart.
- r_2 is more negative than for the other lags because troughs tend to be two quarters behind peaks.
- The dashed blue lines indicate whether the correlations are significantly different from zero

Autocorrelation functions

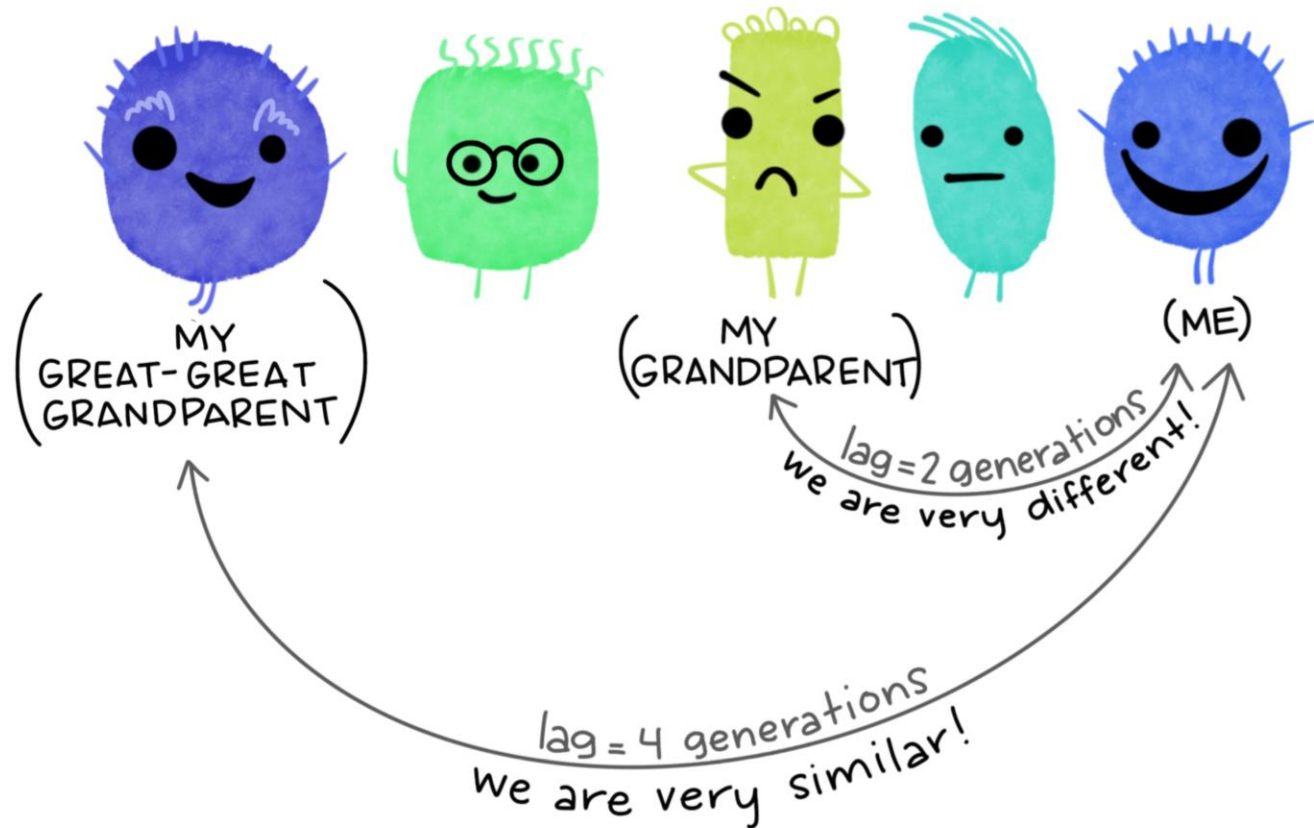
intro to the autocorrelation function (ACF)



in our family MONSTERS tend to be...

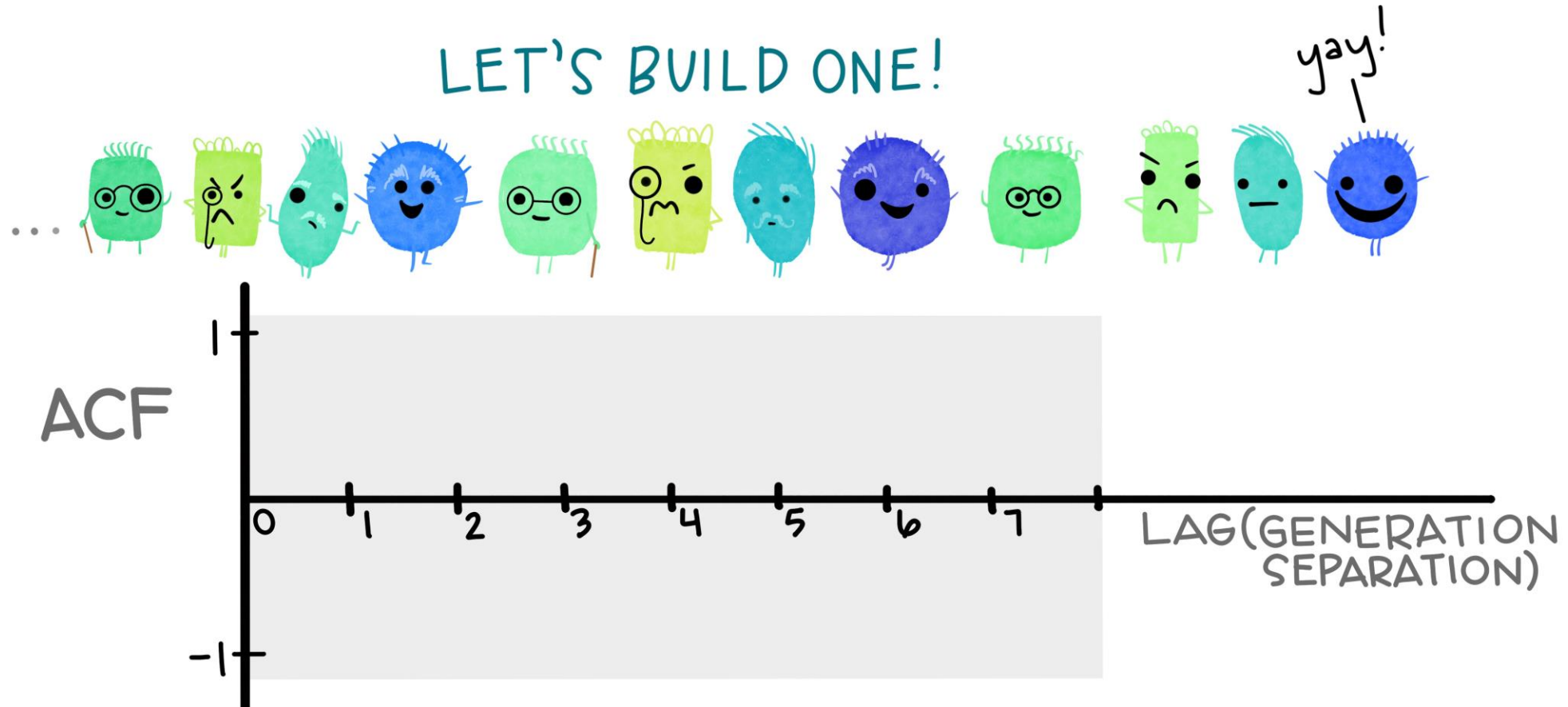
- A little similar to their parent and great-grandparent
- Very different from their grandparent
- Very similar to their great-great grandparent

FOR EXAMPLE:



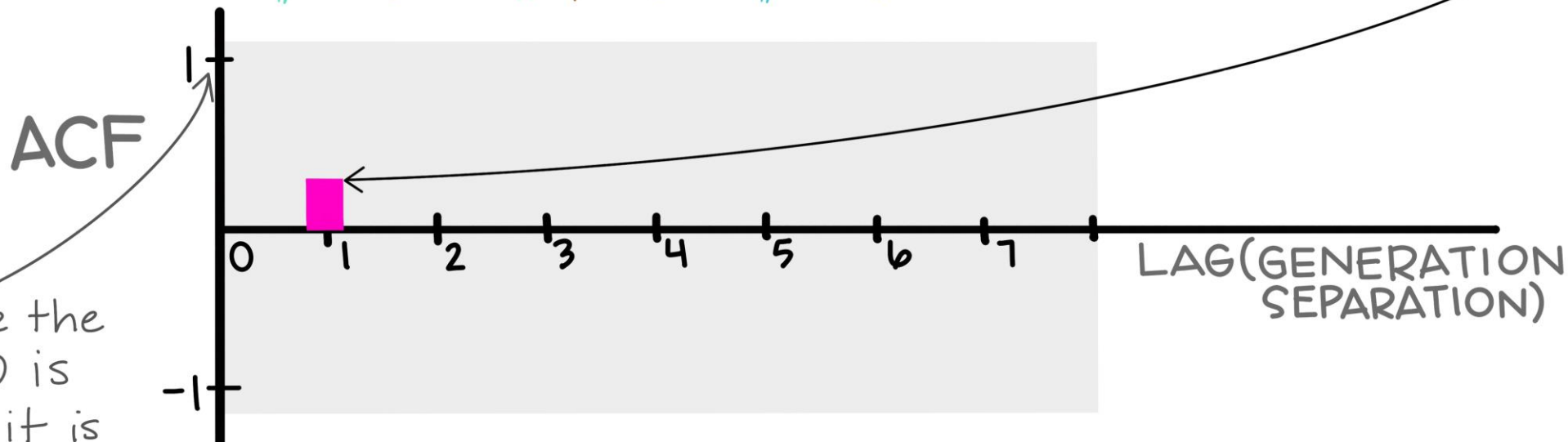
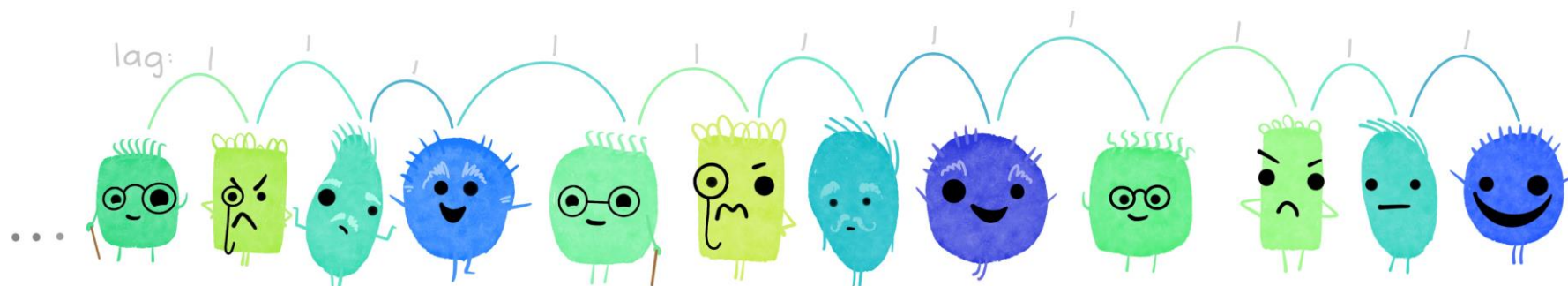
THE autocorrelation function (ACF)

The ACF is a plot of autocorrelation between a variable and itself separated by specified lags (in our case, generations)



At lag = 1, we find the correlation between
monsters and their **parent**.

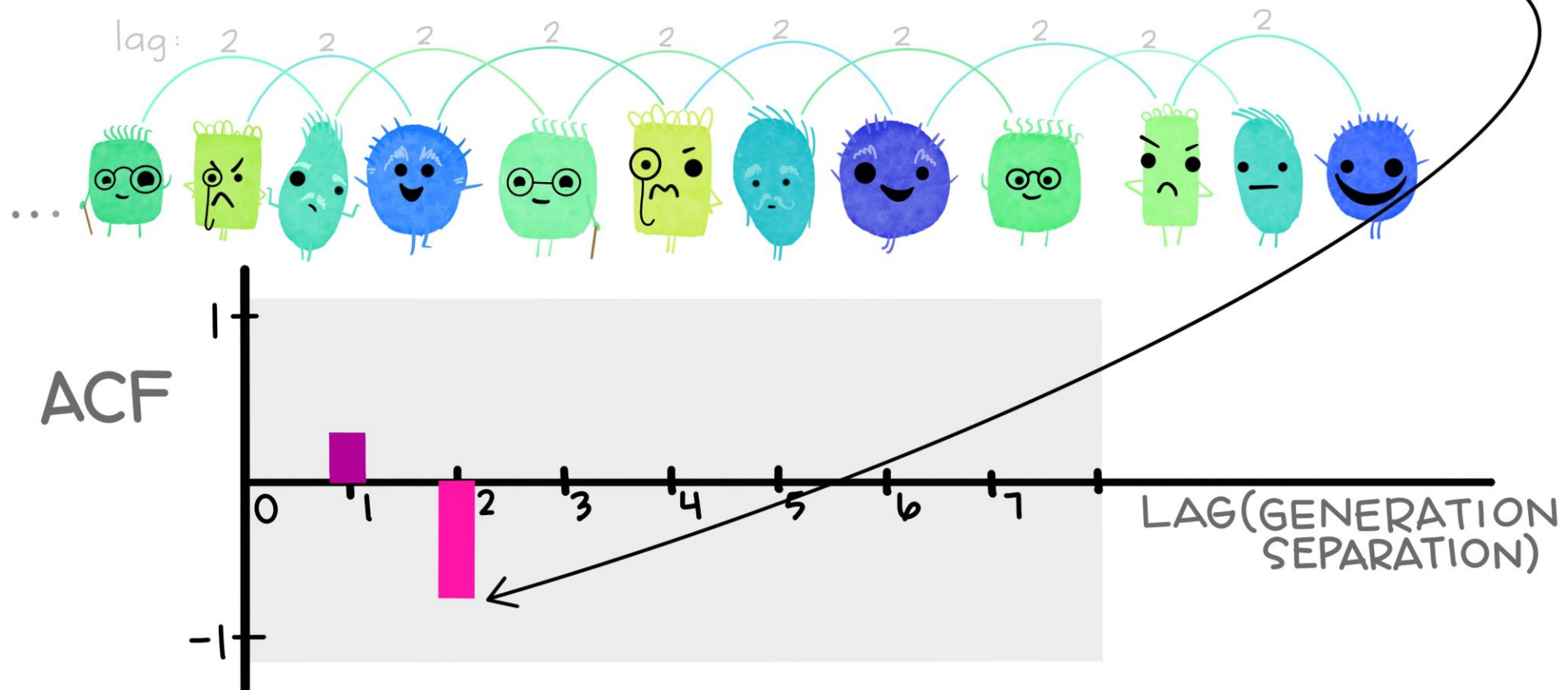
They are somewhat positively correlated.



NOTE: Since the ACF @ lag=0 is always = 1 it is

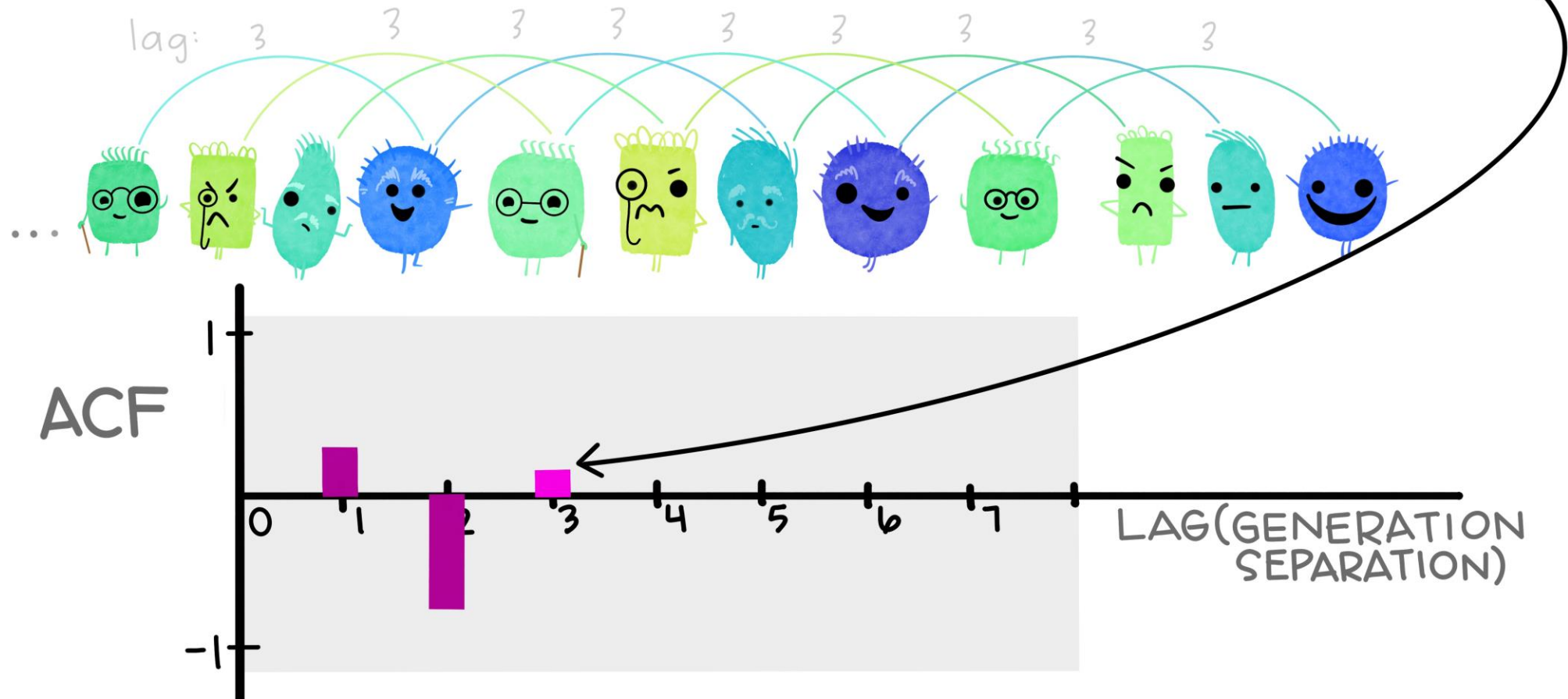
At lag = 2, we find the correlation between **monsters** and their **grandparent**.

Since they tend to be very different, we find a **negative correlation at lag = 2**.



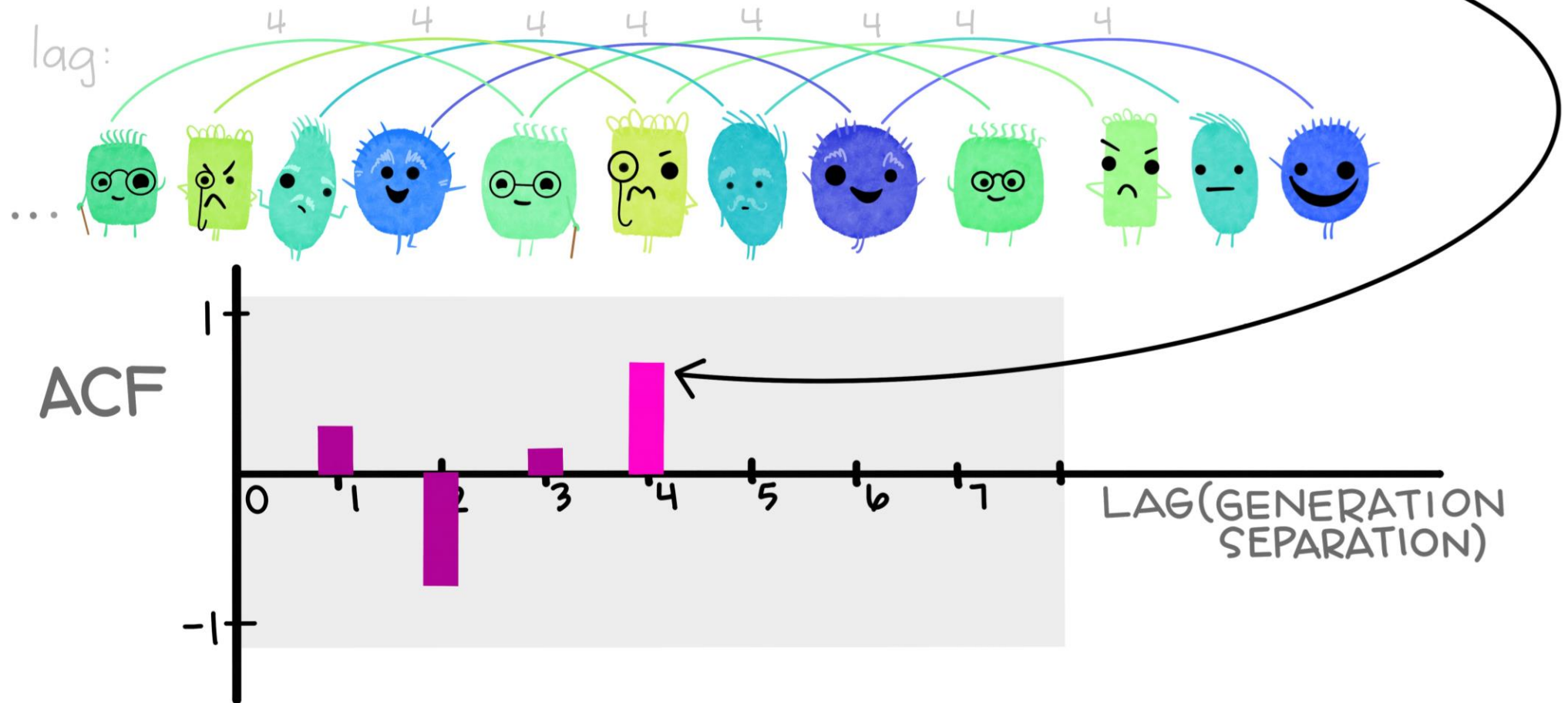
At lag = 3, we find the correlation between **monsters** and their **great-grandparent**.

They are slightly positively correlated.

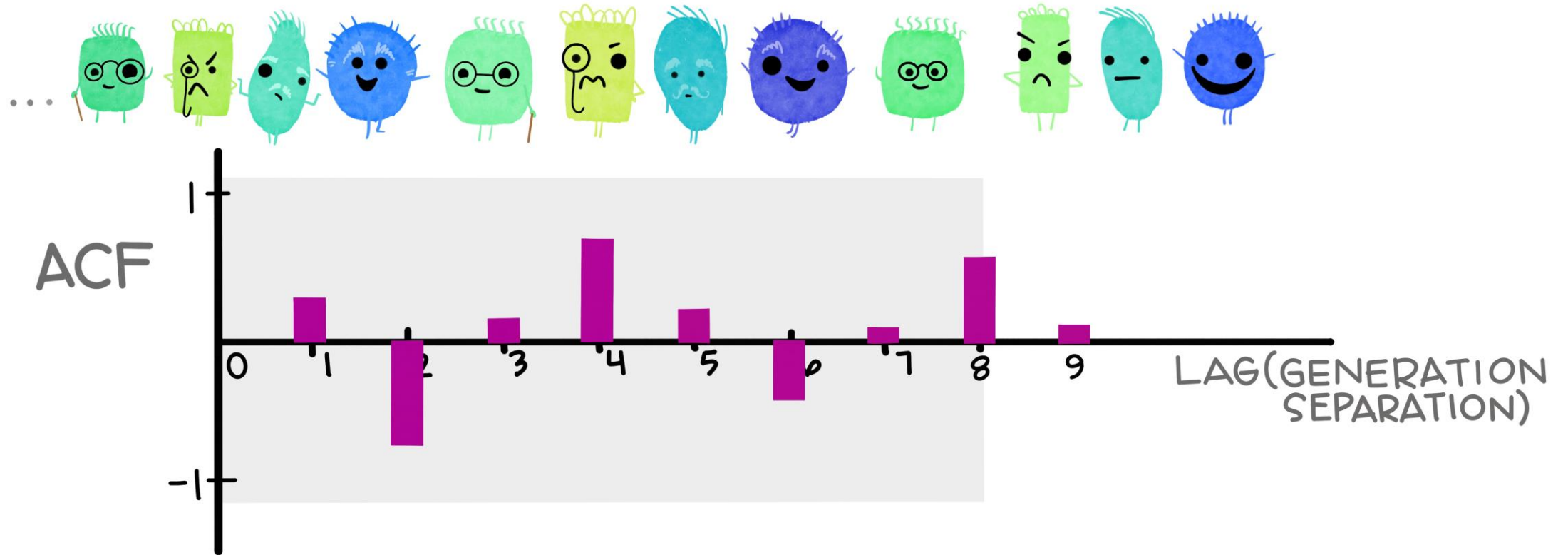


At lag = 4, we find the correlation between **monsters** and their **great-great grandparent**.

They tend to be very similar
(there is a positive correlation).

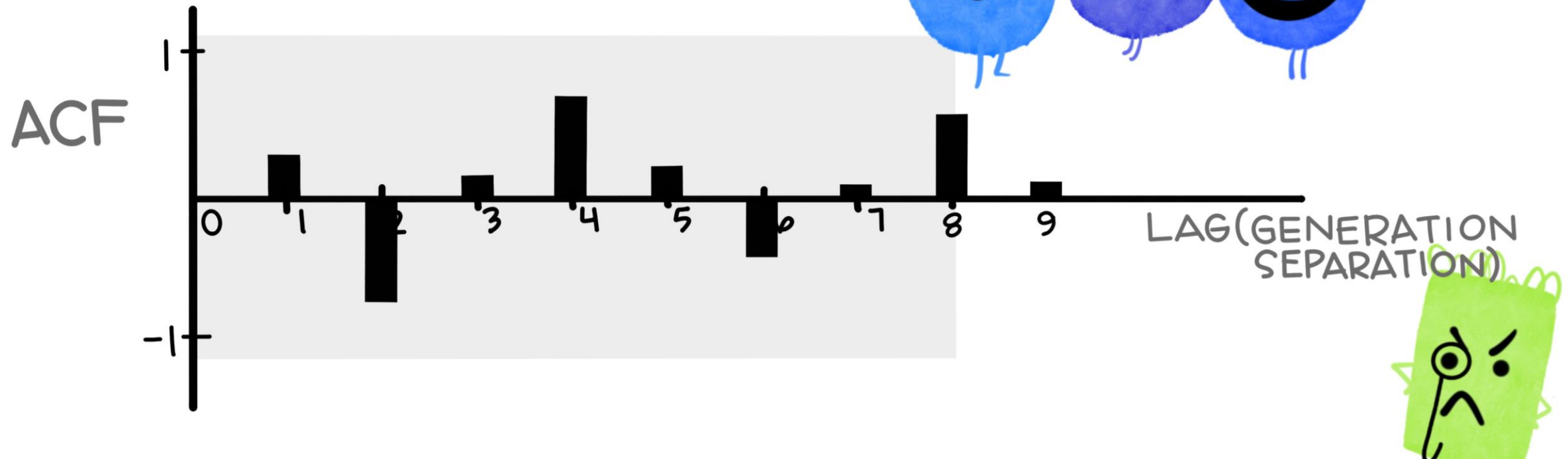


...and we continue finding the correlations as we increase the lag (generations) between the monsters...



in Summary:

The autocorrelation function (ACF) tells us the correlation between observations and those that came before them, separated by different lags (here, monster generations)!



Trend and seasonality in ACF plots

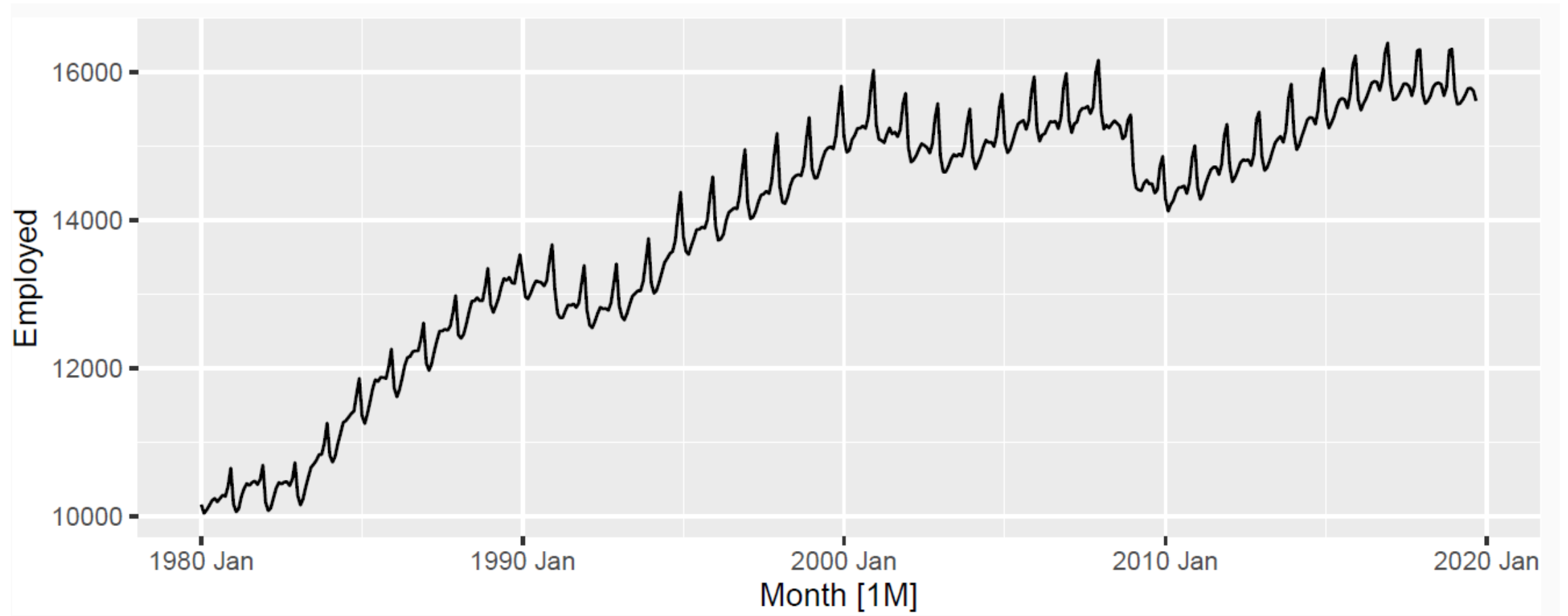
When data have a trend, the autocorrelations for small lags tend to be large and positive because observations nearby in time are also nearby in value.

So, the ACF of a trended time series tends to have positive values that slowly decrease as the lags increase.

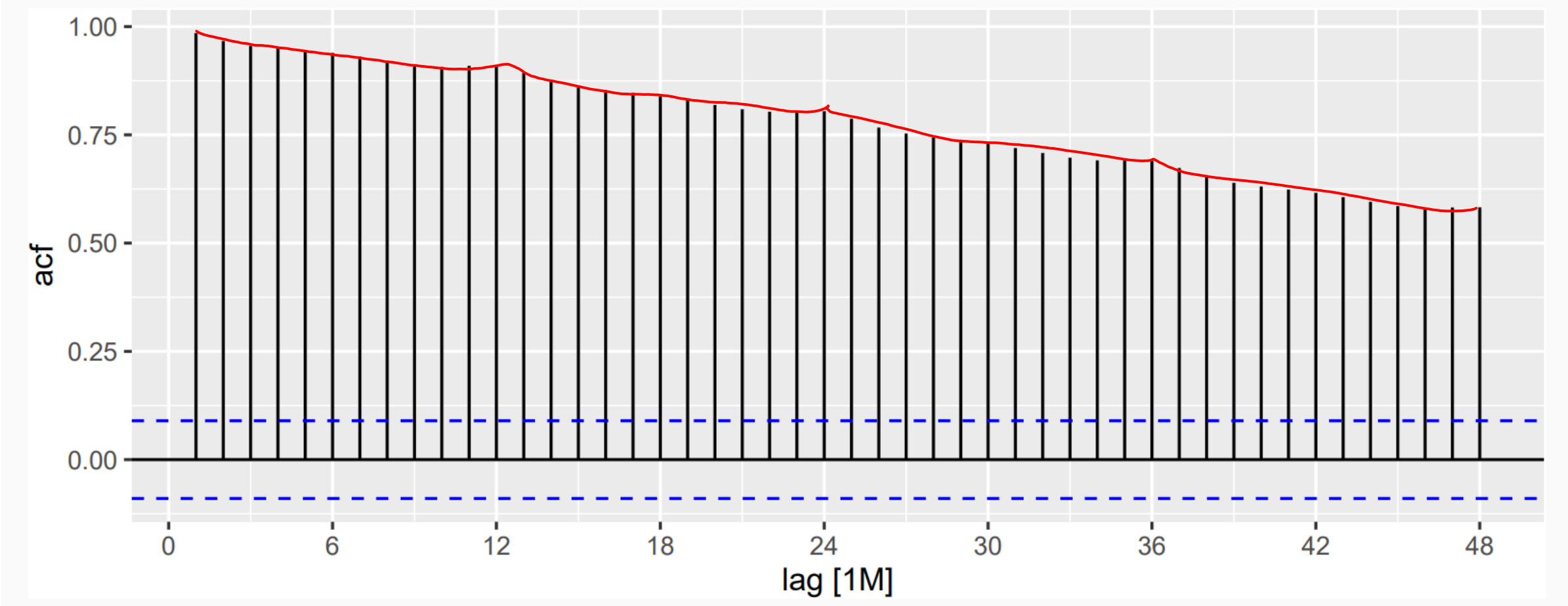
When data are seasonal, the autocorrelations will be larger for the seasonal lags (at multiples of the seasonal period) than for other lags.

When data are both trended and seasonal, we see a combination of these effects.

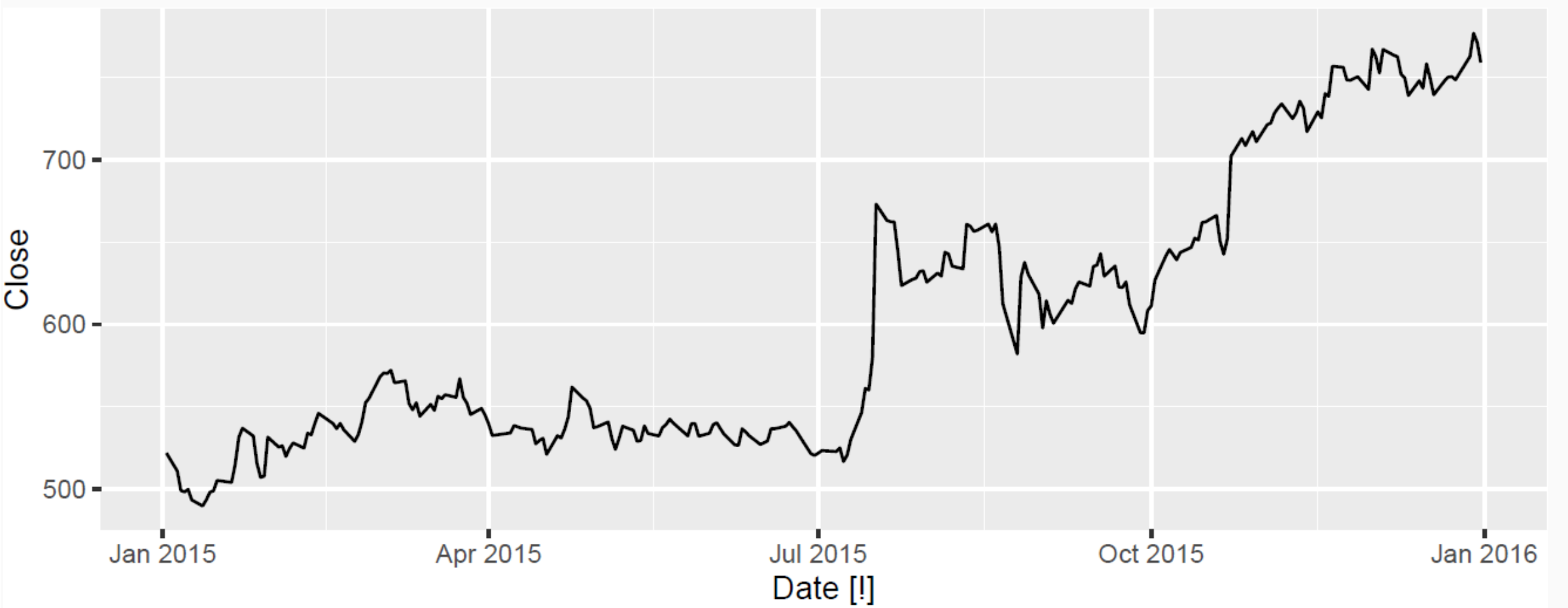
US retail trade employment



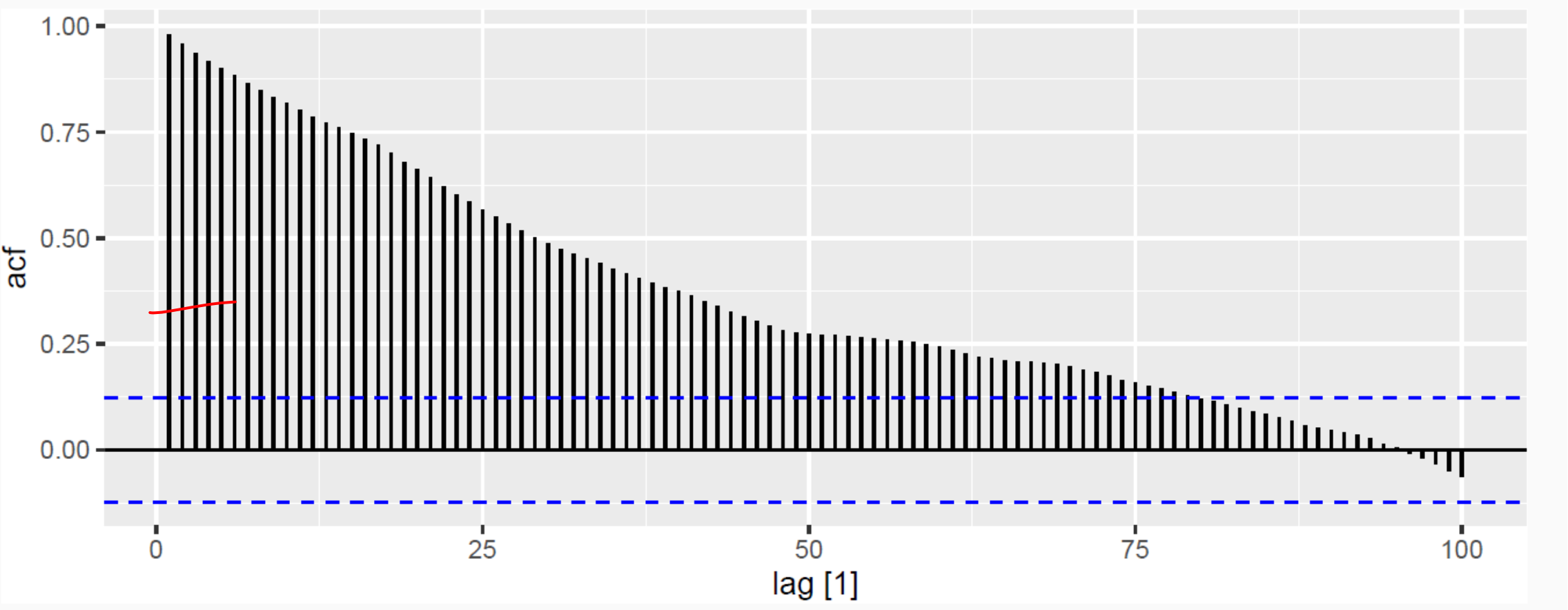
Trend and seasonality in ACF plots



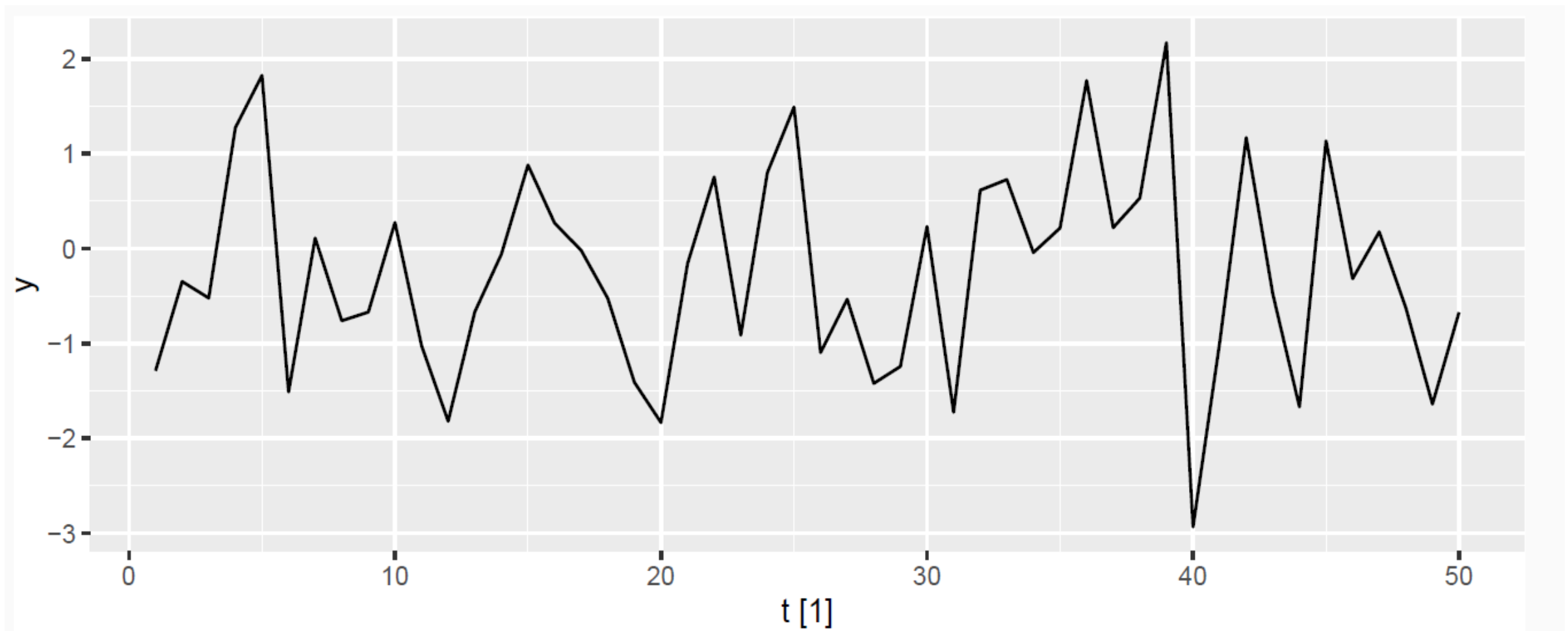
Google stock price



Google stock price ACF plot



White noise



White noise

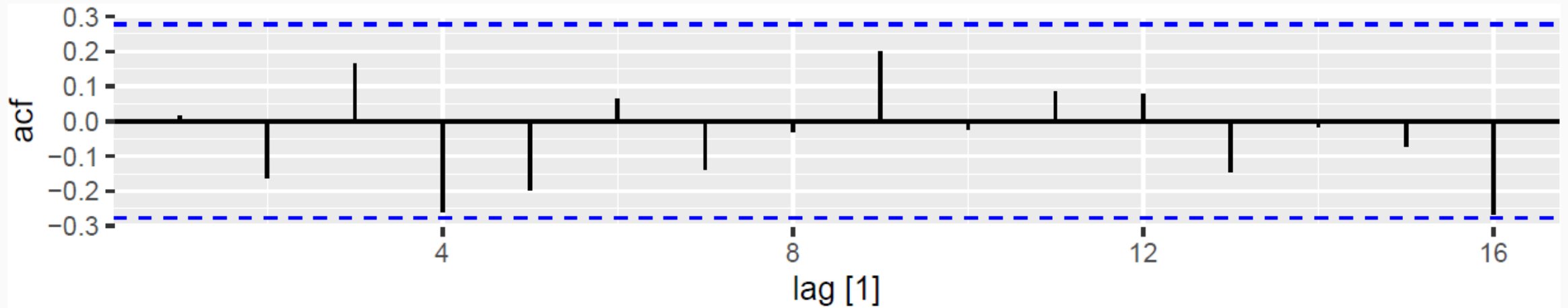
$$\text{VAR}(w_n) = 2t^2 \quad \times, \quad \text{VAR}(w_n) = 2 \quad \checkmark$$



White noise data is uncorrelated across time with zero mean and constant variance.
(Technically, we require independence as well.)

White noise ACF PLOT

r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9	r_{10}
0.014	-0.163	0.163	-0.259	-0.198	0.064	-0.139	-0.032	0.199	-0.024



- Sample autocorrelations for white noise series.
- Expect each autocorrelation to be close to zero.
- Blue lines show 95% critical values.

Sampling distribution of autocorrelations

$$\text{S.E.} = \sigma / \sqrt{n} = \text{Stdev}$$

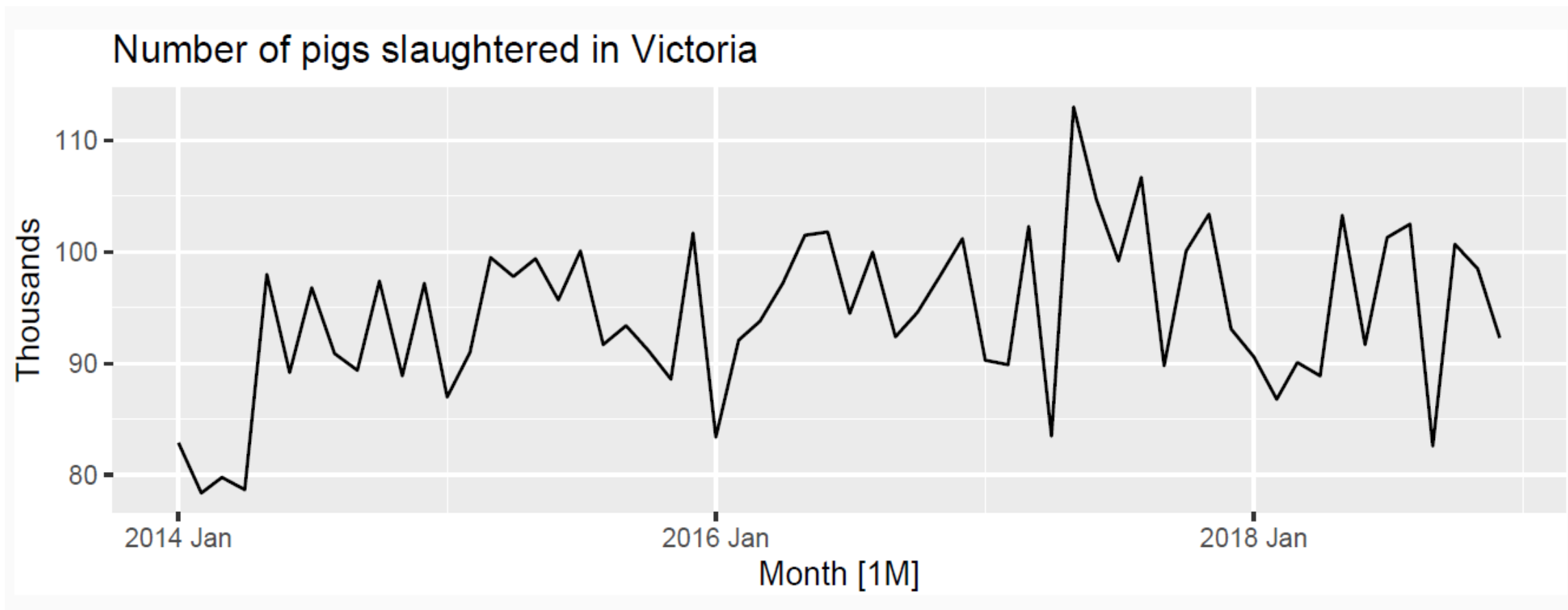
Sampling distribution of r_k for white noise data is asymptotically $N(0, 1/T)$.

$$N(\mu, \sigma^2)$$

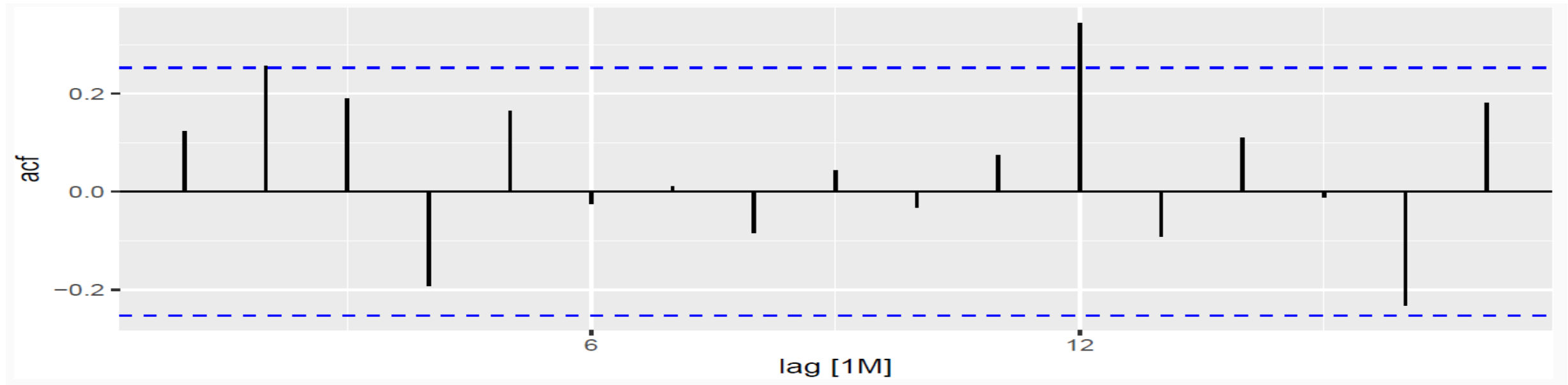
$$\mu \pm Z_{\frac{\alpha}{2}}(\text{S.E.})$$

- 95% of all r_k for white noise must lie within $\pm 1.96 / \sqrt{T}$. ✓
- If this is not the case, the series is probably not WN.
- Common to plot lines at $\pm 1.96 / \sqrt{T}$ when plotting ACF. These are the **critical values**. ✓

Pigs slaughtered



Pigs slaughtered



Monthly total number of pigs slaughtered in the state of Victoria, Australia, from January 2014 through December 2018 (Source:Australian Bureau of Statistics.)

Difficult to detect pattern in time plot.

ACF shows significant autocorrelation for lag 2 and 12.

Indicate some slight seasonality.

These show the series is **not a white noise series**.

Recall

Trend pattern exists when there is a long-term increase or decrease in the data.

Cyclic pattern exists when data exhibit rises and falls that are *not of fixed period* (duration usually of at least 2 years).

Seasonal pattern exists when a series is influenced by seasonal factors (e.g., the quarter of the year, the month, or day of the week).

Time series decomposition

$$y_t = f(S_t, T_t, R_t)$$

where y_t = data at period t

T_t = trend-cycle component at period t

S_t = seasonal component at period t

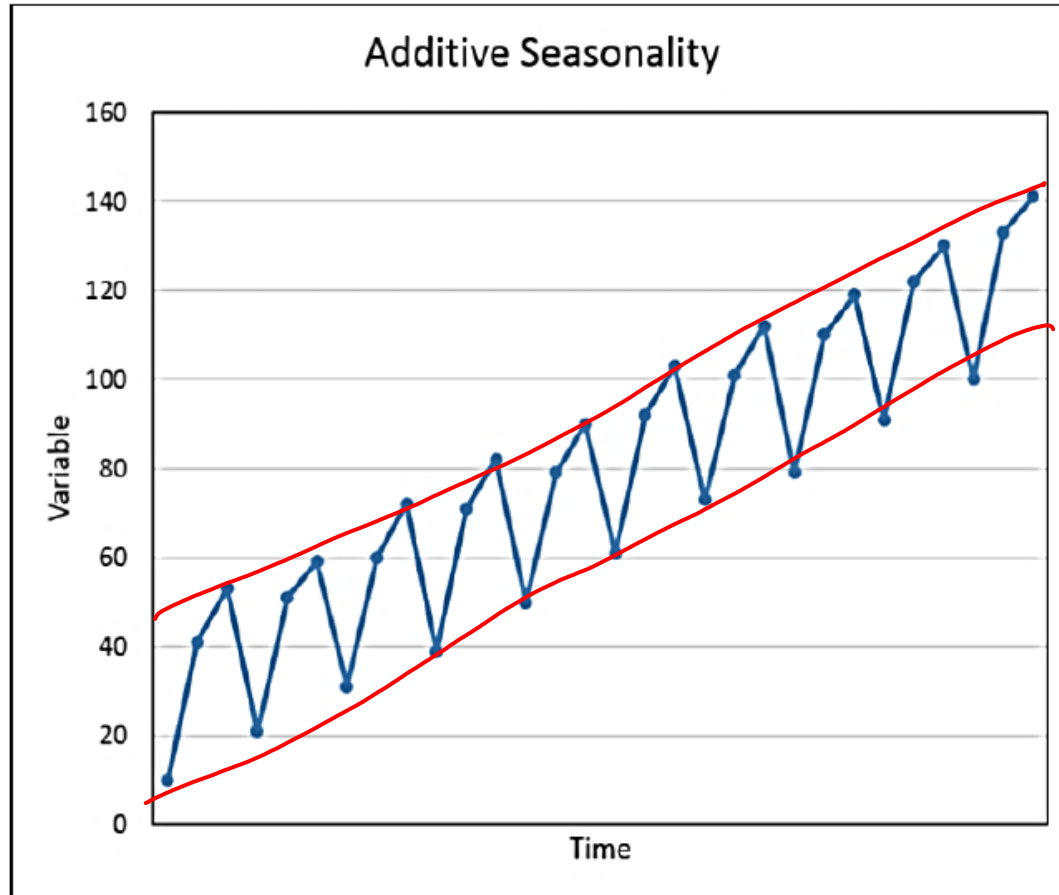
R_t = remainder component at period t

Additive decomposition: $y_t = S_t + T_t + R_t$.

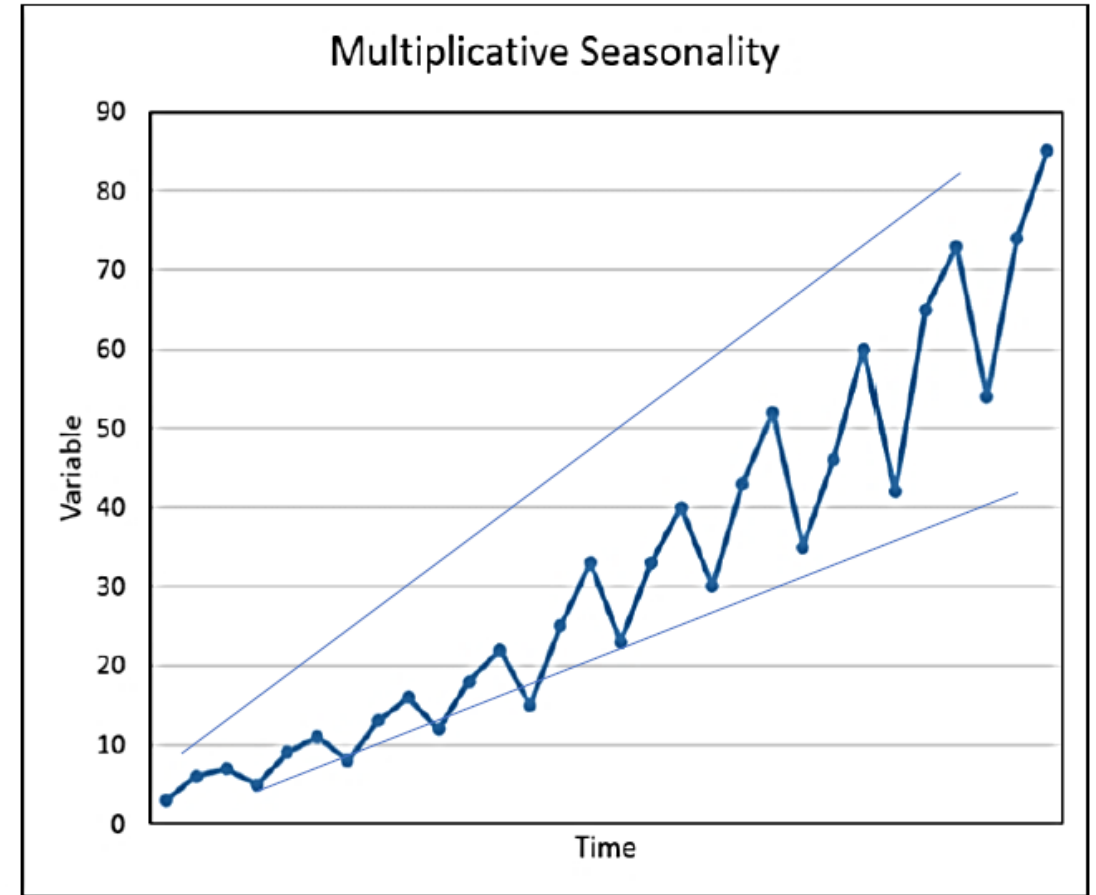
Multiplicative decomposition: $y_t = S_t \times T_t \times R_t$.

$$\log y_t = \log S_t + \log T_t + \log R_t$$

Additive vs Multiplicative Seasonality



Magnitude of the seasonal patterns remains the same with time or trend. Height of the peak and the troughs remain more or less the same.



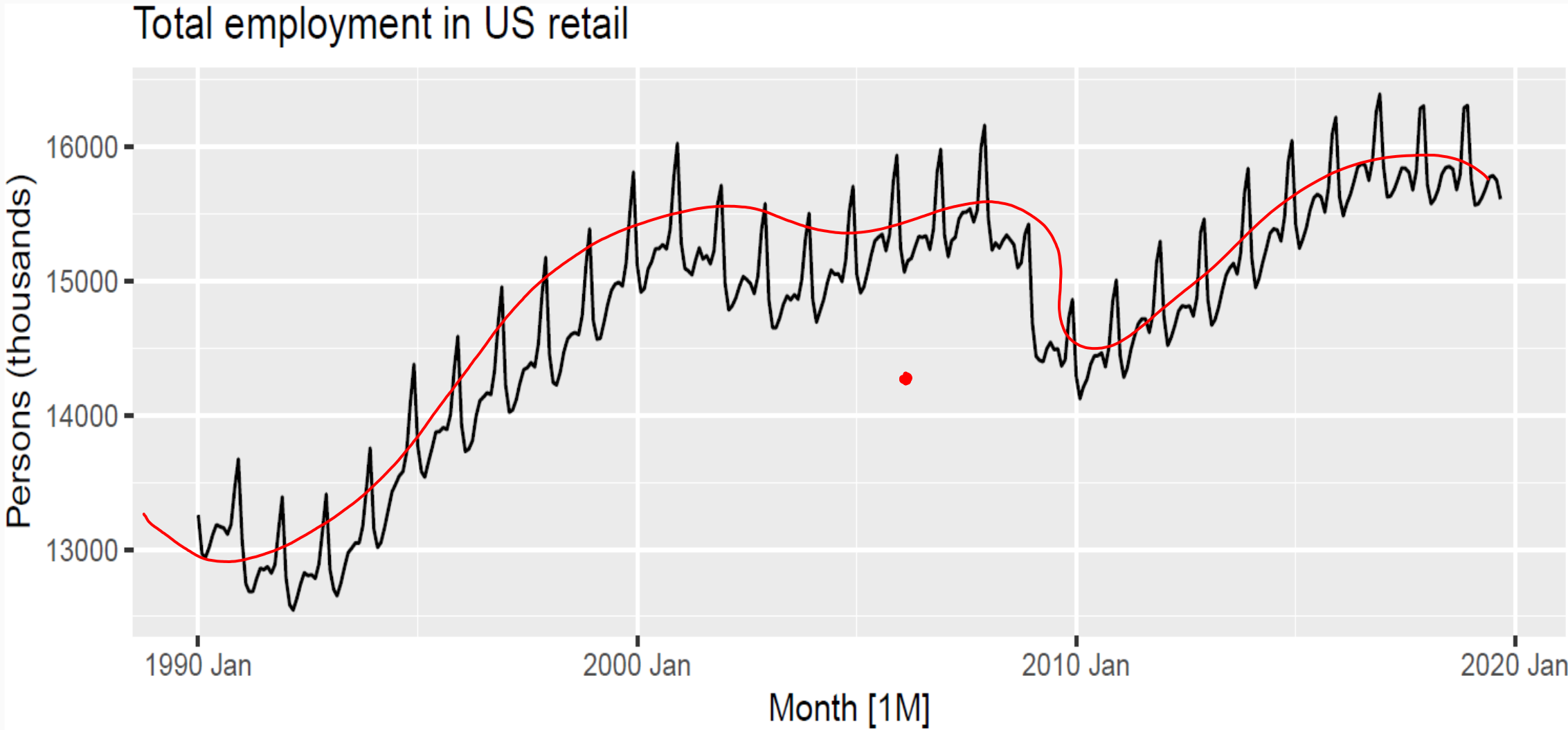
Seasonal behaviour that changes with the time. With increase in trend, magnitude increases. Magnitude of seasonal variation falls with the falling trend.

Time series decomposition

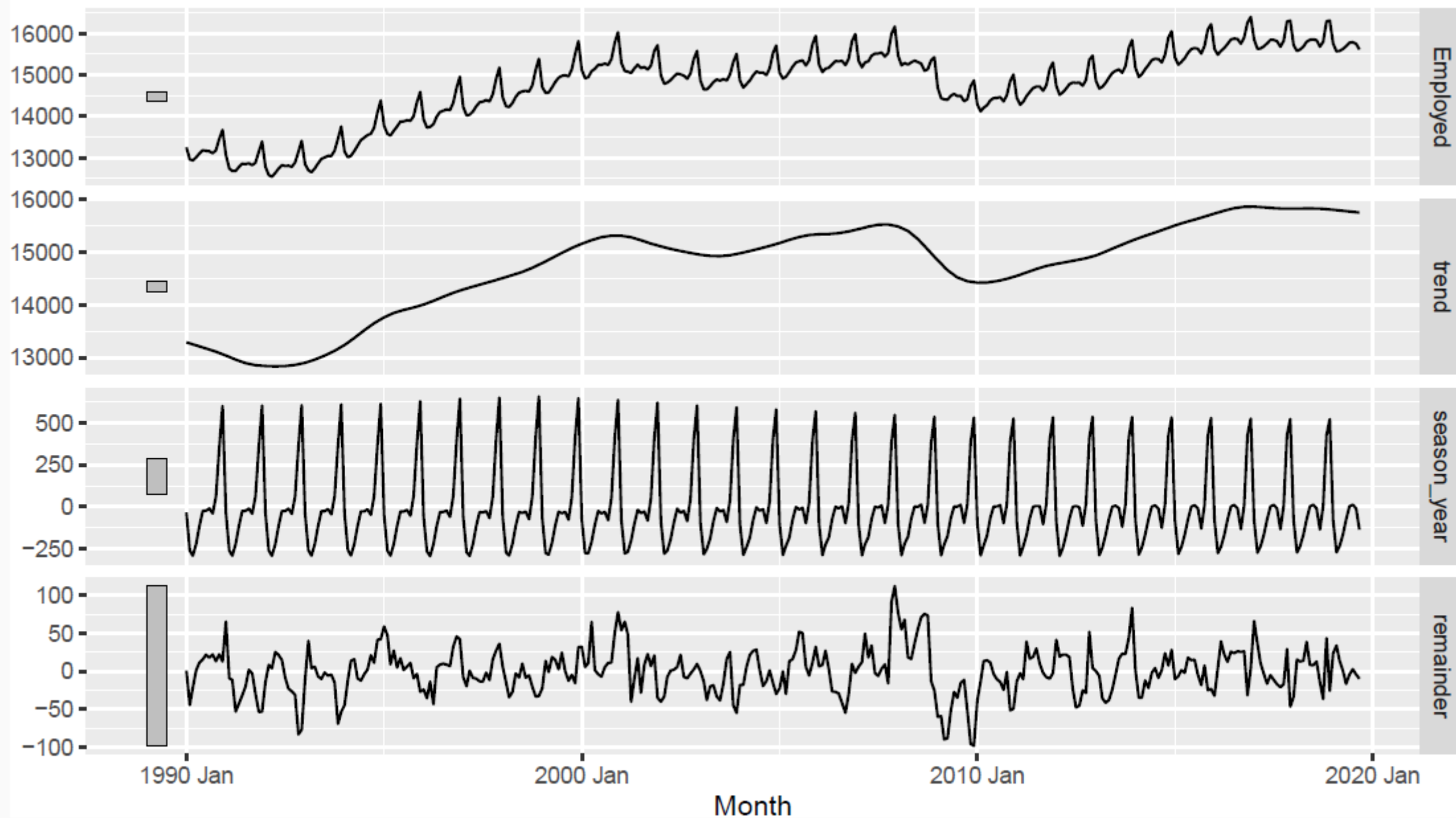
- Additive model appropriate if magnitude of seasonal fluctuations does not vary with level.
- If seasonal are proportional to level of series, then multiplicative model appropriate.
- Multiplicative decomposition more prevalent with economic series
- Alternative: use a Box-Cox transformation, and then use additive decomposition.
- Logs turn multiplicative relationship into an additive relationship:

$$y_t = S_t \times T_t \times R_t \quad \Rightarrow \quad \log y_t = \log S_t + \log T_t + \log R_t.$$

US Retail Employment Time Plot

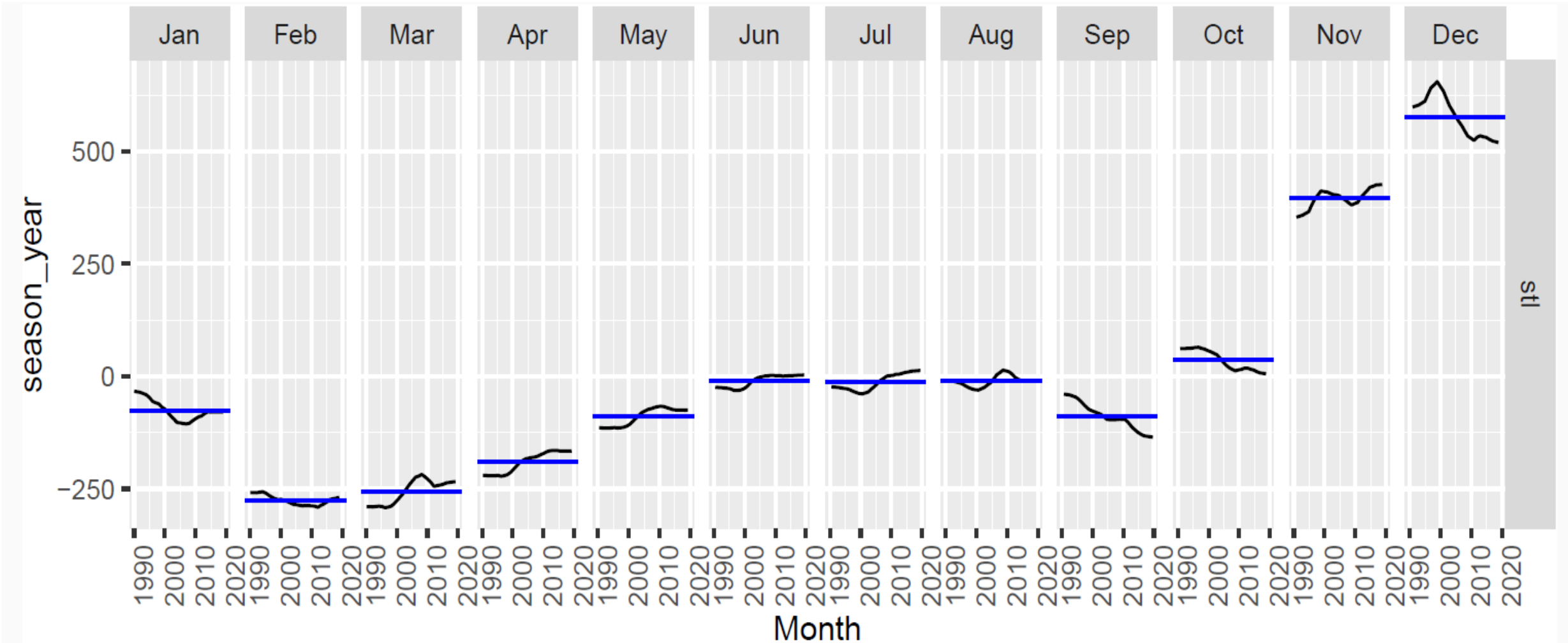


Decomposition



US Retail Employment

Sub Series Plot



Seasonal adjustment

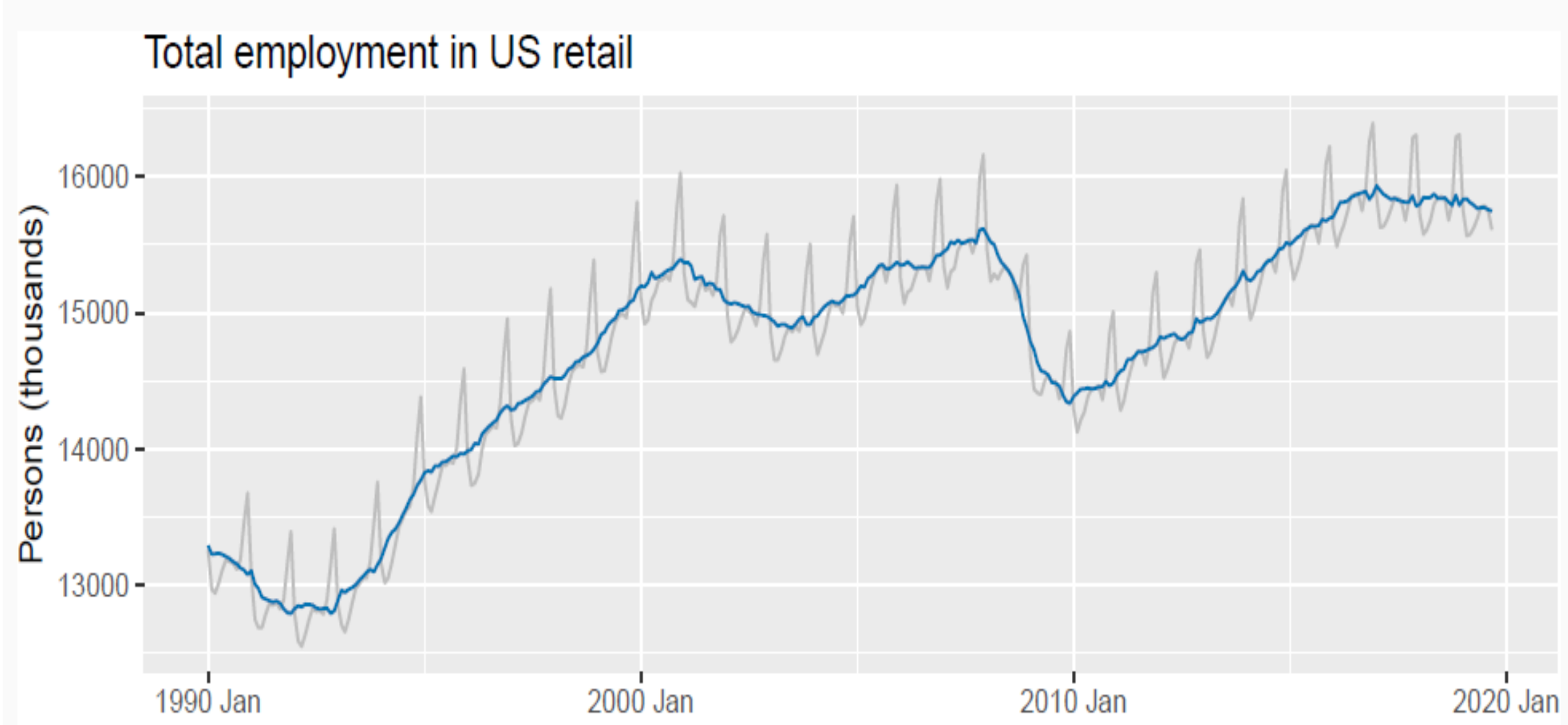
- Useful by-product of decomposition: an easy way to calculate seasonally adjusted data.
- Additive decomposition: seasonally adjusted data given by

$$y_t - \underline{S_t} = T_t + R_t$$

- Multiplicative decomposition: seasonally adjusted data given by

$$y_t / \underline{S_t} = T_t \times R_t$$

US Retail Employment



Seasonal adjustment

- We use estimates of S based on past values to seasonally adjust a current value.
- Seasonally adjusted series reflect **remainders** as well as **trend**.
- Therefore they are not “smooth” and “downturns” or “upturns” can be misleading.
- It is better to use the trend-cycle component to look for turning points.

Moving averages

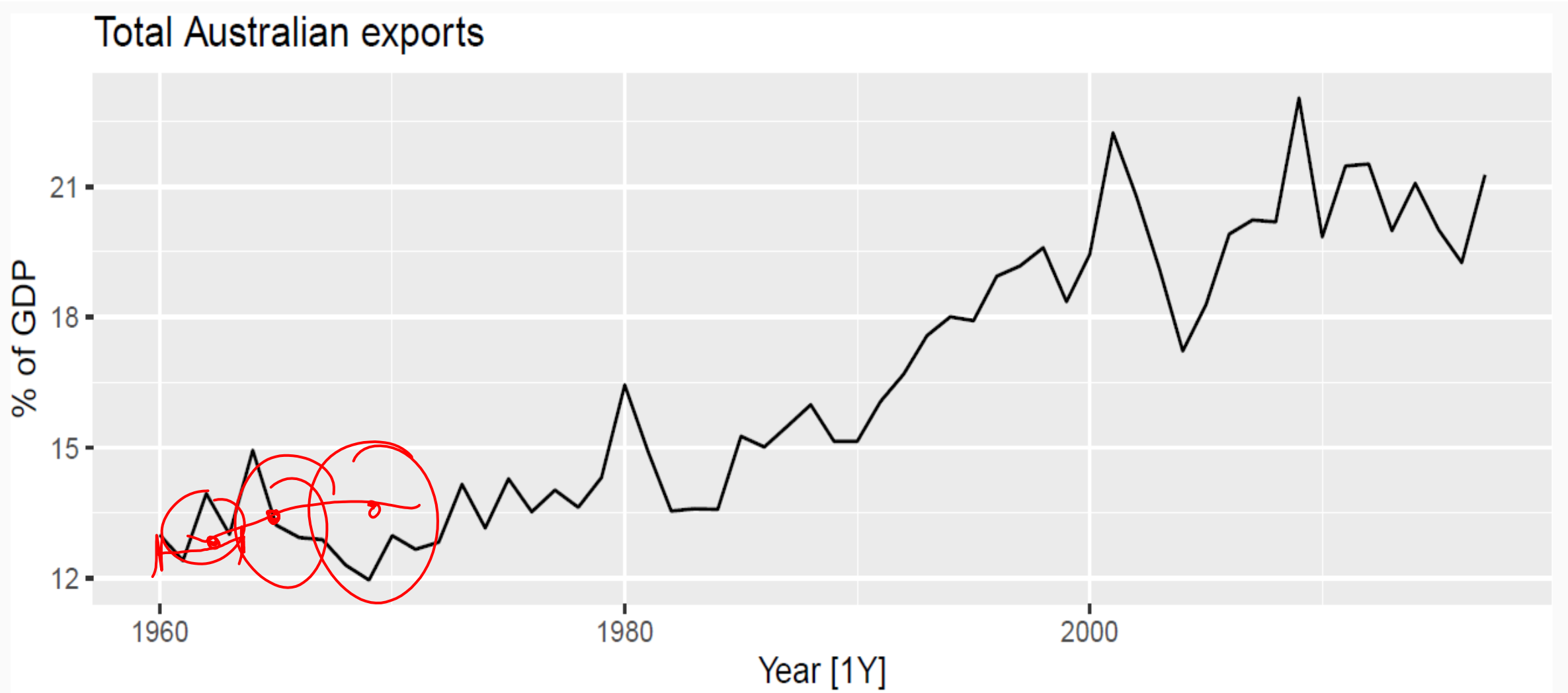
The simplest estimate of the trend-cycle uses **moving averages**.

m-MA

$$\hat{T}_t = \frac{1}{m} \sum_{j=-k}^k y_{t+j}$$

where $k = \frac{m-1}{2}$.

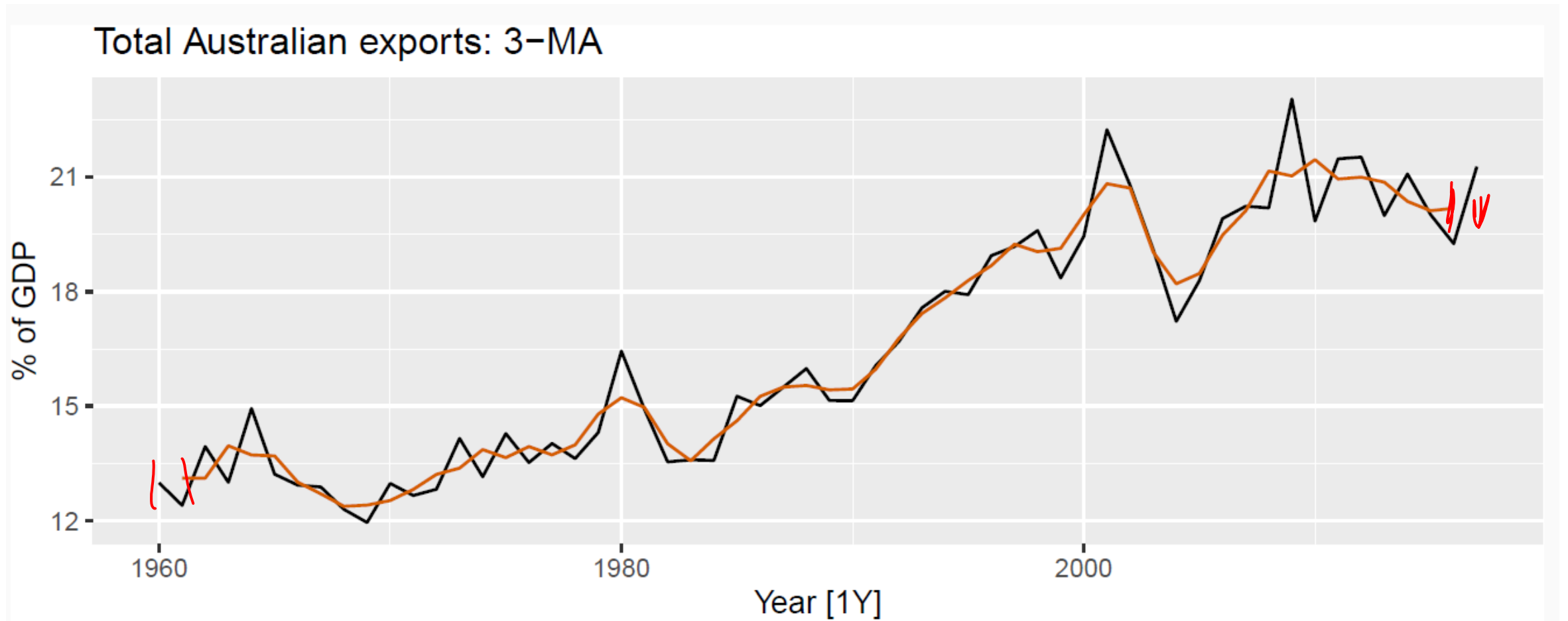
Moving averages



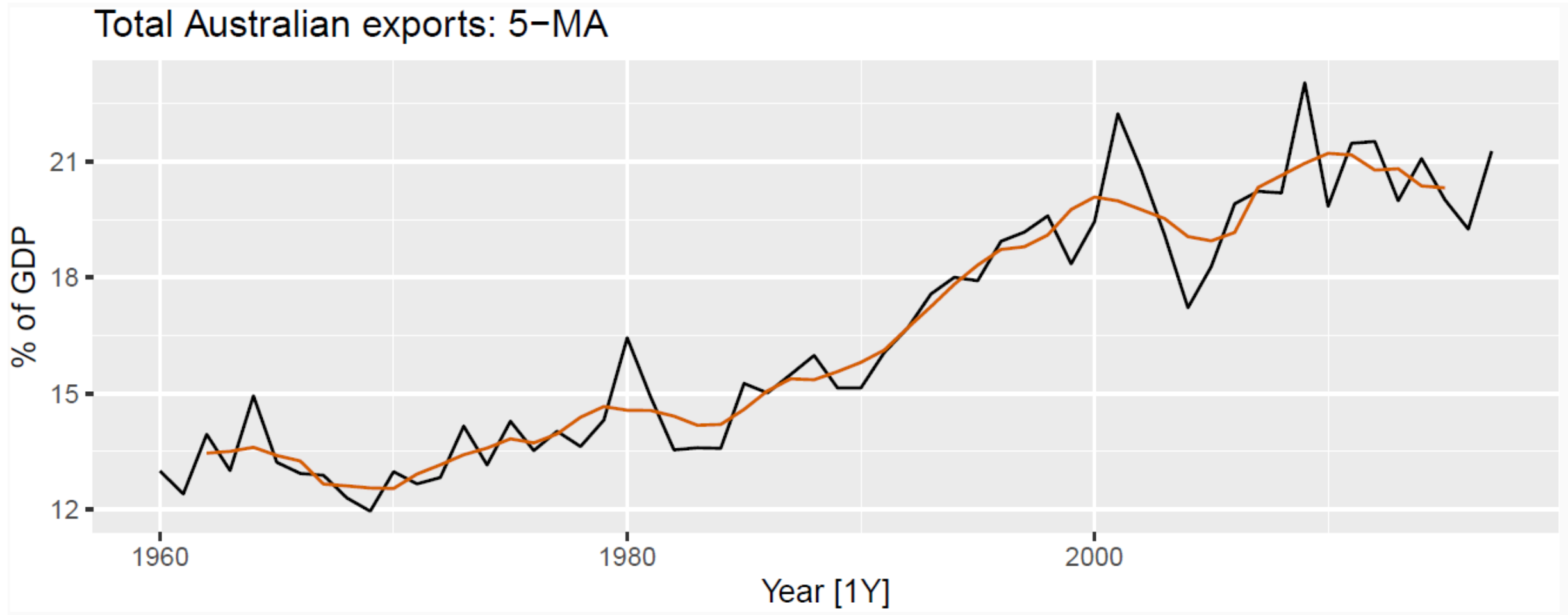
Moving average smoothing

Year	Exports	5-MA
1960.00	12.99	
1961.00	12.40	
1962.00	13.94	13.46
1963.00	13.01	13.50
1964.00	14.94	13.61
...	...	...
2012.00	21.52	20.78
2013.00	19.99	20.81
2014.00	21.08	20.37
2015.00	20.01	20.32
2016.00	19.25	
2017.00	21.27	

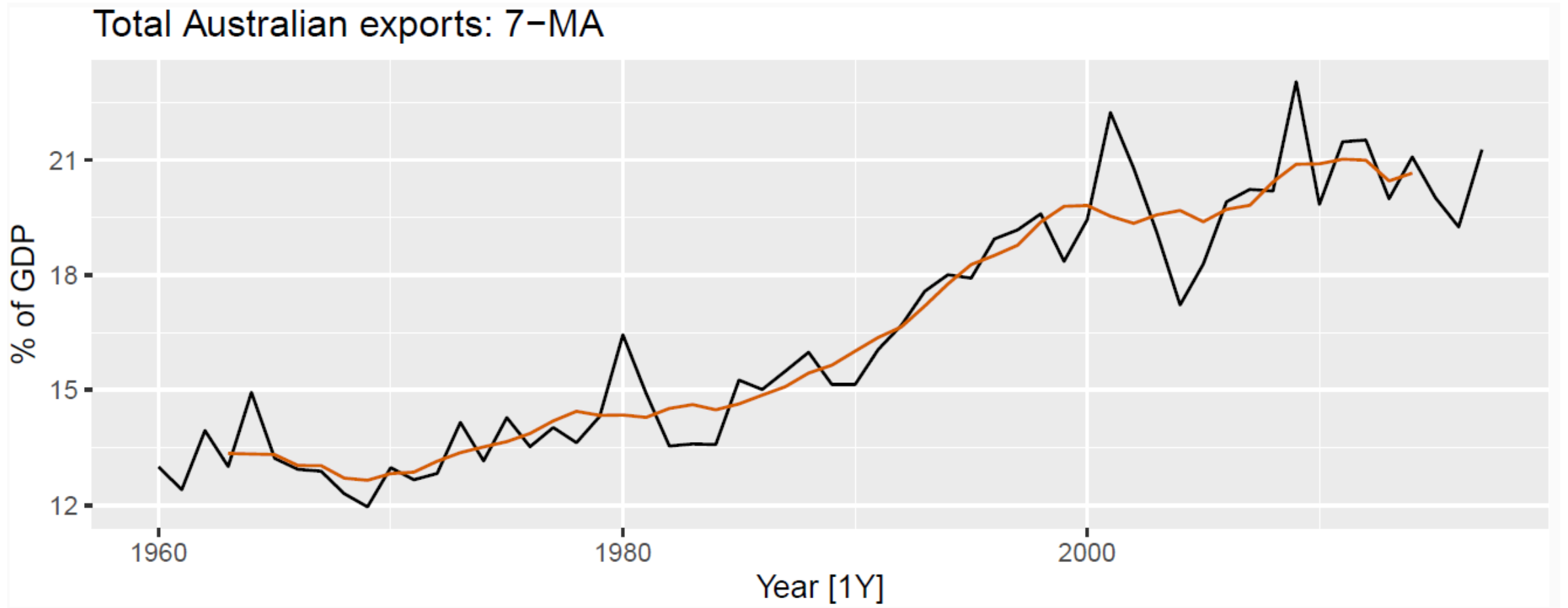
Moving average smoothing



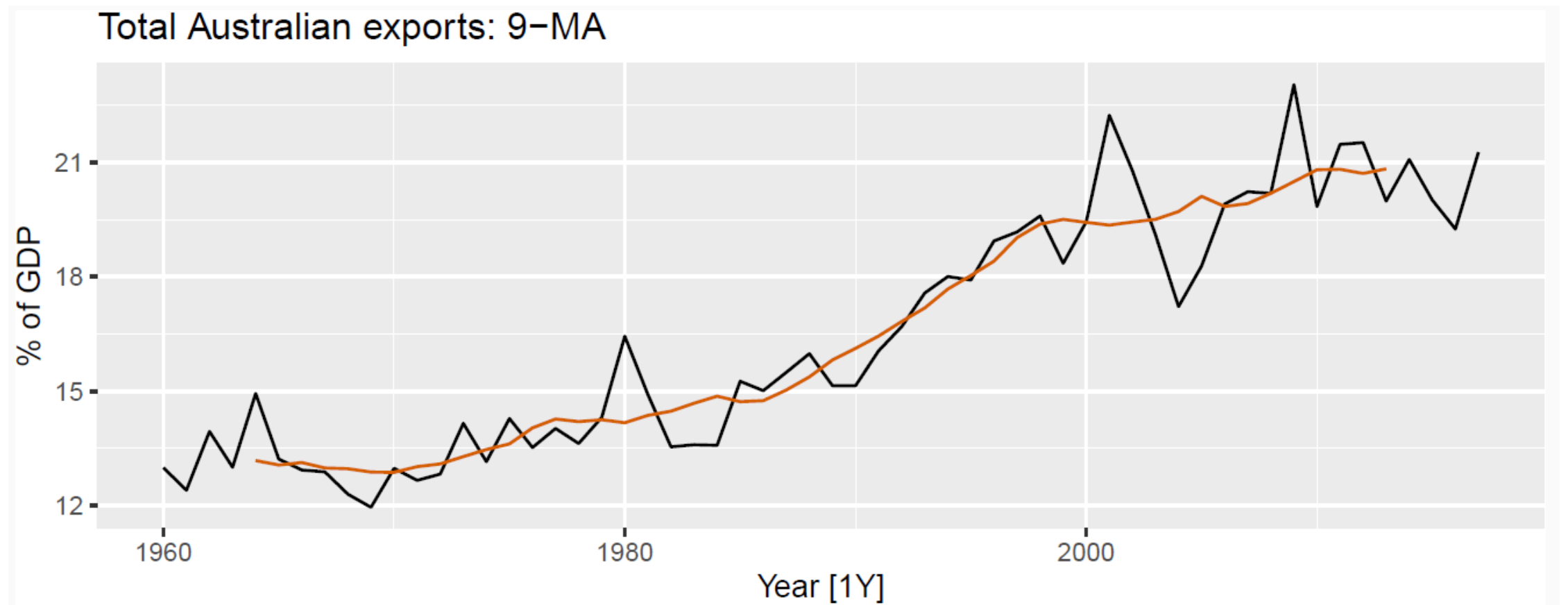
Moving average smoothing



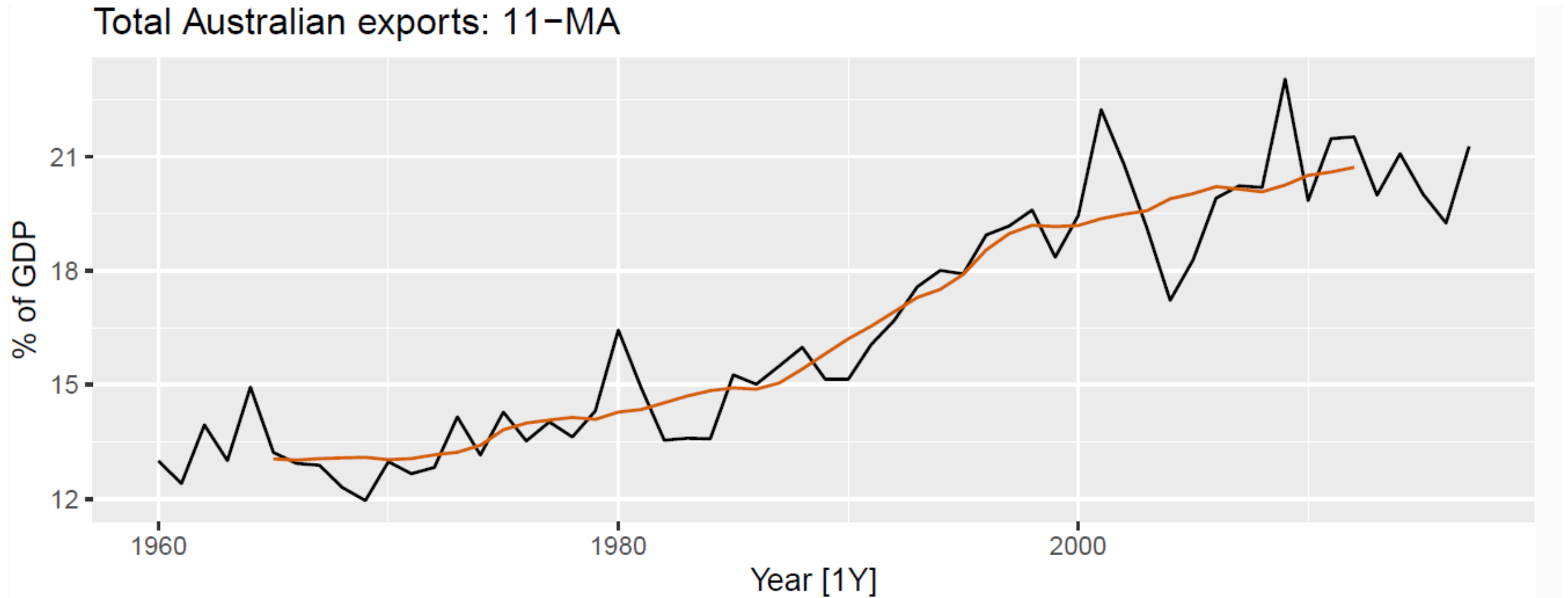
Moving average smoothing



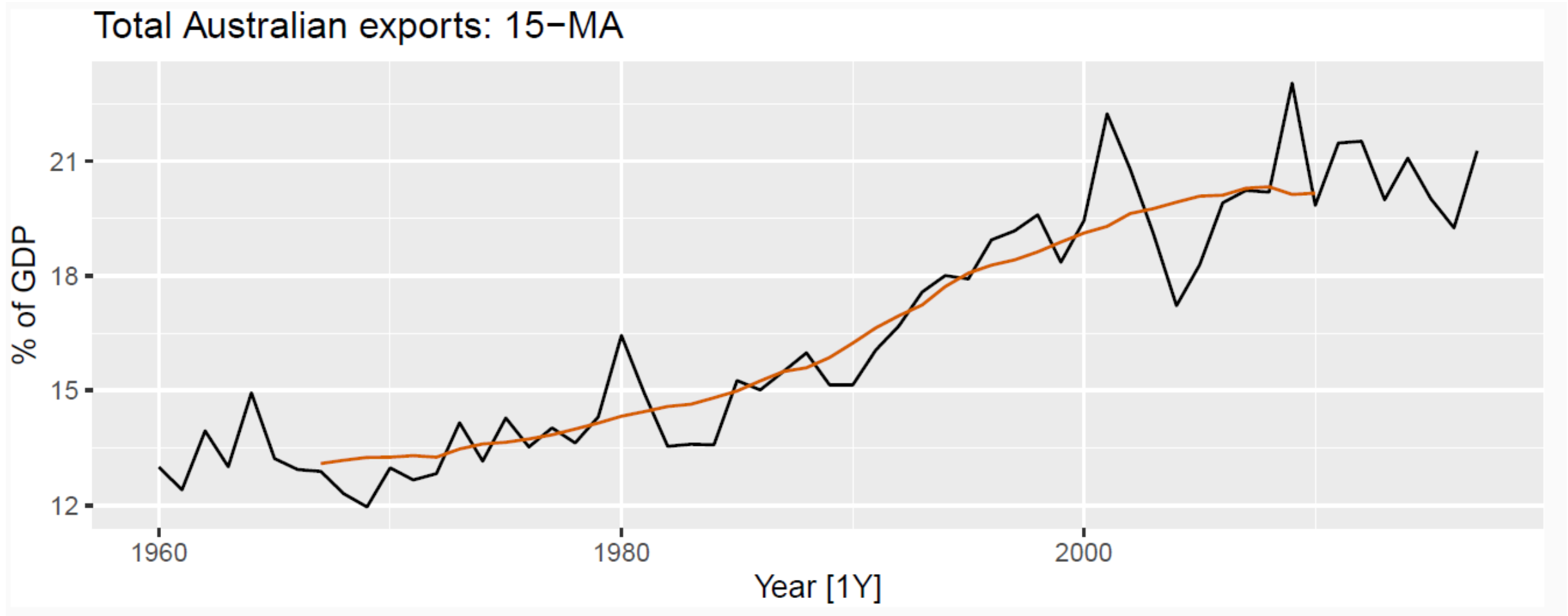
Moving average smoothing



Moving average smoothing



Moving average smoothing



Moving average smoothing

So a moving average is an **average of nearby points**

- observations nearby in time are also likely to be **close in value**.
- average eliminates some **randomness** in the data, leaving a smooth trend-cycle component.

$$\text{3-MA: } \hat{T}_t = (y_{t-1} + y_t + y_{t+1})/3$$

$$\text{5-MA: } \hat{T}_t = (y_{t-2} + y_{t-1} + y_t + y_{t+1} + y_{t+2})/5$$

- each average computed by dropping **oldest** observation and including **next** observation.
- averaging **moves** through time series until trend-cycle computed at each observation possible.

Why is there no estimate at ends?

- For a 3 MA, there cannot be estimates at time 1 or time T because the observations at time 0 and $T + 1$ are not available.
- Generally: there cannot be estimates at times near the endpoints.

The order of the MA

- larger order means smoother, flatter curve
- larger order means more points lost at ends
- order = length of season or cycle removes pattern
- But so far odd orders?

Cantered MA

4 MA:

$$\frac{1}{4}(y_{t-2} + y_{t-1} + y_t + y_{t+1})$$

or

$$\frac{1}{4}(y_{t-1} + y_t + y_{t+1} + y_{t+2})$$

Cantered MA

4 MA:

$$\frac{1}{4}(y_{t-2} + y_{t-1} + y_t + y_{t+1})$$

or
$$\frac{1}{4}(y_{t-1} + y_t + y_{t+1} + y_{t+2})$$

Solution: take a further 2-MA to “centre” result.

4 MA

= 2

$$\sum_{k=-2} a_{ii} y_{t-k}$$

$$\begin{aligned} T_t &= \frac{1}{2} \left\{ \frac{1}{4}(y_{t-2} + y_{t-1} + y_t + y_{t+1}) \right. \\ &\quad \left. + \frac{1}{4}(y_{t-1} + y_t + y_{t+1} + y_{t+2}) \right\} \\ &= \frac{1}{8}y_{t-2} + \frac{1}{4}y_{t-1} + \frac{1}{4}y_t + \frac{1}{4}y_{t+1} + \frac{1}{8}y_{t+2} \end{aligned}$$

Cantered MA

Year	Data	4-MA	$2 \times 4\text{-MA}$
1992 Q1	443.00		
1992 Q2	410.00	451.25	
1992 Q3	420.00	448.75	450.00
1992 Q4	532.00	451.50	450.12
1993 Q1	433.00	449.00	450.25
1993 Q2	421.00	444.00	446.50
1993 Q3	410.00	448.00	446.00
1993 Q4	512.00	438.00	443.00
⋮	⋮	⋮	⋮

Moving average trend-cycle

A moving average of the same length as the season removes the seasonal pattern.

- For quarterly data: use a 2×4 MA
- For monthly data: use a 2×12 MA

$$\hat{T}_t = \frac{1}{24}y_{t-6} + \frac{1}{12}y_{t-5} + \dots + \frac{1}{12}y_{t+5} + \frac{1}{24}y_{t+6}$$

Time series decomposition

Additive decomposition

Step 1

If m is an even number, compute the trend-cycle component $\hat{T}_t$ using a $2 \times m$ -MA. If m is an odd number, compute the trend-cycle component $\hat{T}_t$ using an m -MA.

Step 2

Calculate the detrended series: $y_t - \hat{T}_t$.

Step 3

To estimate the seasonal component for each season, simply average the detrended values for that season. For example, with monthly data, the seasonal component for March is the average of all the detrended March values in the data. These seasonal component values are then adjusted to ensure that they add to zero. The seasonal component is obtained by stringing together these monthly values, and then replicating the sequence for each year of data. This gives $\hat{S}_t$.

Step 4

The remainder component is calculated by subtracting the estimated seasonal and trend-cycle components: $\hat{R}_t = y_t - \hat{T}_t - \hat{S}_t$.

Multiplicative decomposition

A classical multiplicative decomposition is similar, except that the subtractions are replaced by divisions.

Step 1

If m is an even number, compute the trend-cycle component $\hat{T}_t$ using a $2 \times m$ -MA. If m is an odd number, compute the trend-cycle component $\hat{T}_t$ using an m -MA.

Step 2

Calculate the detrended series: $y_t/\hat{T}_t$.

Step 3

To estimate the seasonal component for each season, simply average the detrended values for that season. For example, with monthly data, the seasonal index for March is the average of all the detrended March values in the data. These seasonal indexes are then adjusted to ensure that they add to m . The seasonal component is obtained by stringing together these monthly indexes, and then replicating the sequence for each year of data. This gives $\hat{S}_t$.

Step 4

The remainder component is calculated by dividing out the estimated seasonal and trend-cycle components: $\hat{R}_t = y_t/(\hat{T}_t\hat{S}_t)$.

Time series decomposition

Additive decomposition: $y_t = T_t + S_t + R_t = \hat{T}_t + \hat{S}_t + \hat{R}_t$

Multiplicative decomposition: $y_t = T_t \times S_t \times R_t = \hat{T}_t \times \hat{S}_t \times \hat{R}_t$

Time series decomposition

Additive decomposition: $y_t = T_t + S_t + R_t = \hat{T}_t + \hat{S}_t + \hat{R}_t$

Multiplicative decomposition: $y_t = T_t \times S_t \times R_t = \hat{T}_t \times \hat{S}_t \times \hat{R}_t$

- Estimate $\hat{T}$ using $(2 \times m)$ -MA if m is even. Otherwise, estimate $\hat{T}$ using m -MA

Compute de-trended series

- Additive decomposition: $y_t - \hat{T}_t$
- Multiplicative decomposition: $y_t / \hat{T}_t$

De-trending

Remove smoothed series $\hat{T}_t$ from y_t to leave S_t and R_t .

- Additive model: $y_t - \hat{T}_t = (\hat{T}_t + \hat{S}_t + \hat{R}_t) - \hat{T}_t = \hat{S}_t + \hat{R}_t$
- Multiplicative model: $\frac{y_t}{\hat{T}_t} = \frac{\hat{T}_t \times \hat{S}_t \times \hat{R}_t}{\hat{T}_t} = \hat{S}_t \times \hat{R}_t$

Comments on classical decomposition

While classical decomposition is still widely used, it is not recommended, as there are now several much better methods. Some of the problems with classical decomposition are summarised below.

- The estimate of the trend-cycle is unavailable for the first few and last few observations. For example, if $m=12$, there is no trend-cycle estimate for the first six or the last six observations. Consequently, there is also no estimate of the remainder component for the same time periods.
- The trend-cycle estimate tends to over-smooth rapid rises and falls in the data.
- Classical decomposition methods assume that the seasonal component repeats from year to year. For many series, this is a reasonable assumption, but for some longer series it is not. For example, electricity demand patterns have changed over time as air conditioning has become more widespread. In many locations, the seasonal usage pattern from several decades ago had its maximum demand in winter (due to heating), while the current seasonal pattern has its maximum demand in summer (due to air conditioning). Classical decomposition methods are unable to capture these seasonal changes over time.
- Occasionally, the values of the time series in a small number of periods may be particularly unusual. For example, the monthly air passenger traffic may be affected by an industrial dispute, making the traffic during the dispute different from usual. The classical method is not robust to these kinds of unusual values.

ACF and PACF Plots in Time Series Analysis

Autocorrelation Function (ACF)

In time series analysis, ACF and PACF plots are two of the most important tools for identifying the underlying structure of a time series. The ACF plot shows the correlation of a time series with itself at different lags, while the PACF plot shows the correlation of a time series with itself at different lags, after removing the effects of the previous lags.

The ACF plot is a graphical representation of the correlation of a time series with itself at different lags. The correlation coefficient is a measure of how closely two variables are related. A correlation coefficient of 1 indicates a perfect positive relationship, while a correlation coefficient of -1 indicates a perfect negative relationship. A correlation coefficient of 0 indicates no relationship between the two variables.

The ACF plot can be used to identify the order of an MA model. The order of an MA model is the number of lags that are included in the model. The ACF plot will show spikes at the lags that are included in the model.

Partial Autocorrelation Function (PACF)

The PACF plot is a graphical representation of the correlation of a time series with itself at different lags, after removing the effects of the previous lags. The PACF plot can be used to identify the order of an MA model. The order of an MA model is the number of lags that are included in the model. The PACF plot will show spikes at the lags that are included in the model.

Fundamentals

Auto-Regressive and Moving Average Models

Auto-Regressive (AR) Model

$$\hat{y}_t = \alpha_1 y_{t-1} + \cdots + \alpha_p y_{t-p}$$

The AR model assumes that the current value (y_t) is **dependent on previous values** ($y_{t-1}, y_{t-2}, y_{t-3}, \dots$). Because of this assumption, we can build a **linear** regression model.

To figure out the order of an AR model, you would use the **PACF**.

Moving Average (MA) Model

$$\hat{y}_t = \epsilon_t + \beta_1 \epsilon_{t-1} + \cdots + \beta_q \epsilon_{t-q}$$

The MA model assumes that the current value (y_t) is **dependent on the error terms** including the current error ($\epsilon_t, \epsilon_{t-1}, \epsilon_{t-2}, \epsilon_{t-3}, \dots$). Because error terms are random, there is **no linear** relationship between the current value and the error terms.

To figure out the order of an MA model, you would use the **ACF**.

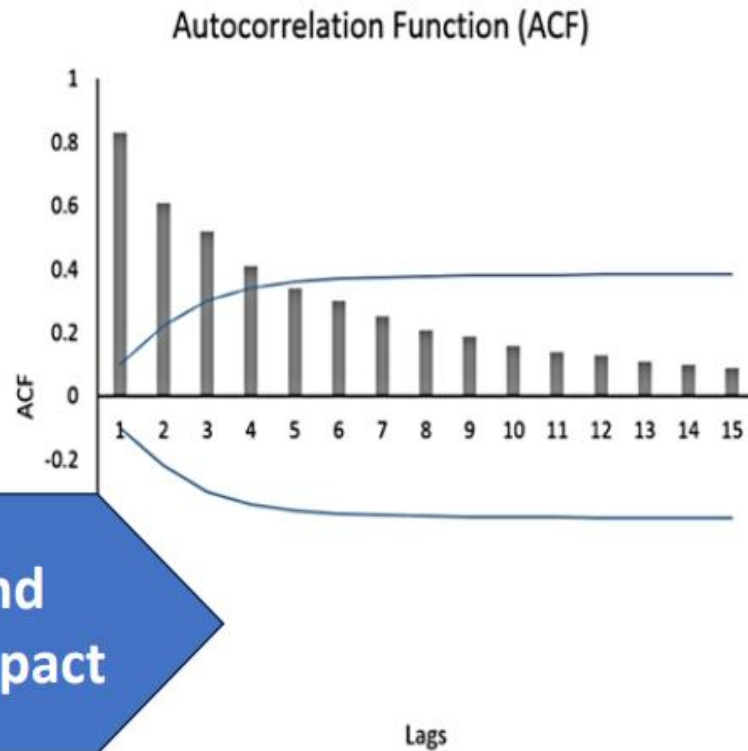
Checking Stationarity

- ACF and PACF assume stationarity of the underlying time series. Stationarity can be checked by performing an **Augmented Dickey-Fuller (ADF) test**:
 - $p\text{-value} > 0.05$: Fail to reject the null hypothesis (H_0), the data has a unit root and is non-stationary.
 - $p\text{-value} \leq 0.05$: Reject the null hypothesis (H_0), the data does not have a unit root and is stationary.
- If the time series is stationary, continue to the next steps. **If the time series is not stationary, try differencing the time series** and check its stationarity again.

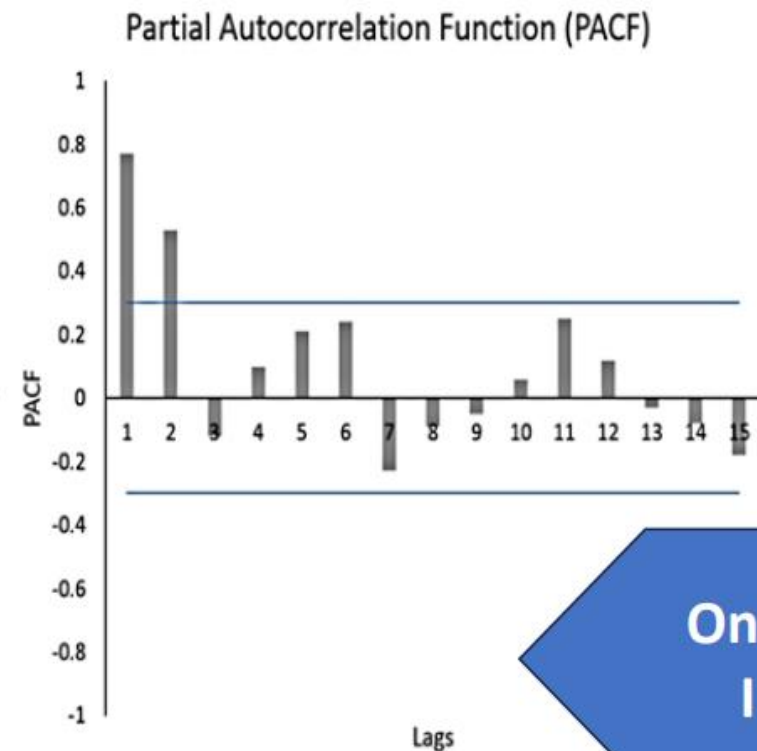
<https://www.kaggle.com/code/iamleonie/time-series-interpreting-acf-and-pacf>

ACF and PACF plots

- When working with time series data, it is often useful to analyze the autocorrelation of the data to understand the patterns and dependencies between time steps.



Direct and
Indirect Impact



Only Direct
Impact

To interpret ACF and PACF plots, we can look for the following patterns:

To determine the order of the model, you can use the following table:

	AR(p)	MA(q)	ARMA(p, q)
ACF	Tails off (Geometric decay)	Significant at lag q / Cuts off after lag q	Tails off (Geometric decay)
PACF	Significant at each lag p / Cuts off after lag p	Tails off (Geometric decay)	Tails off (Geometric decay)

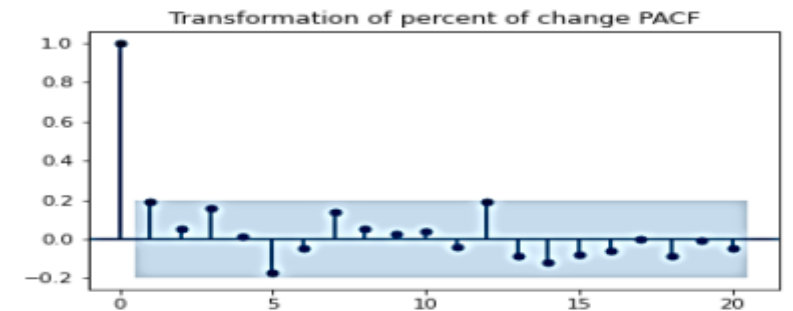
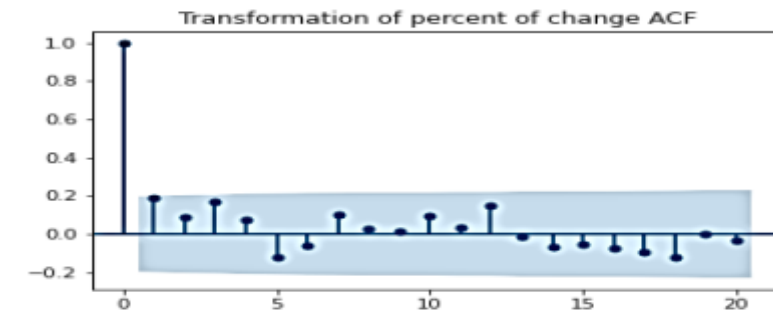
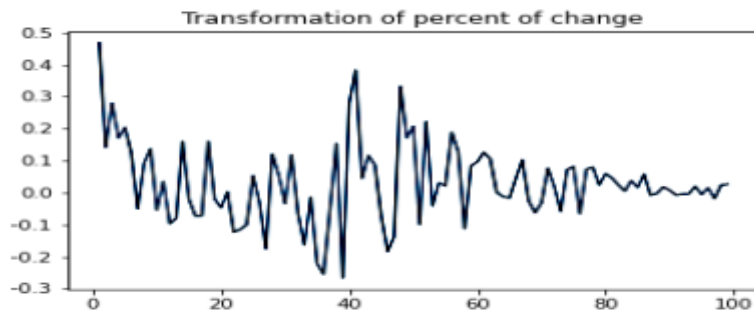
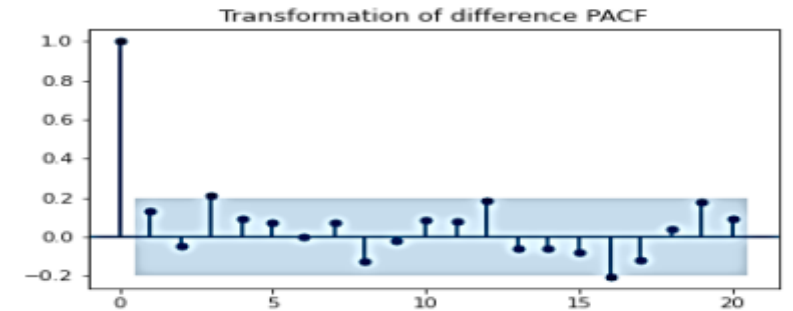
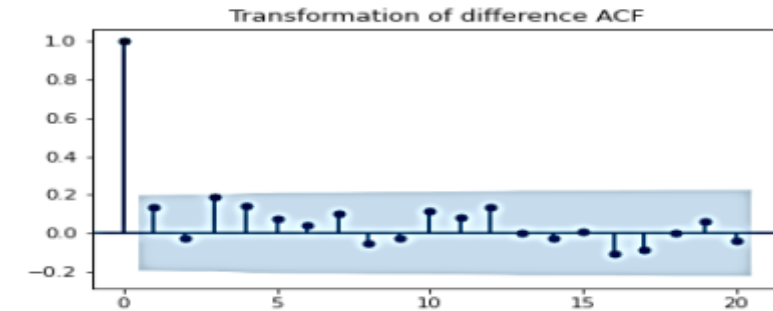
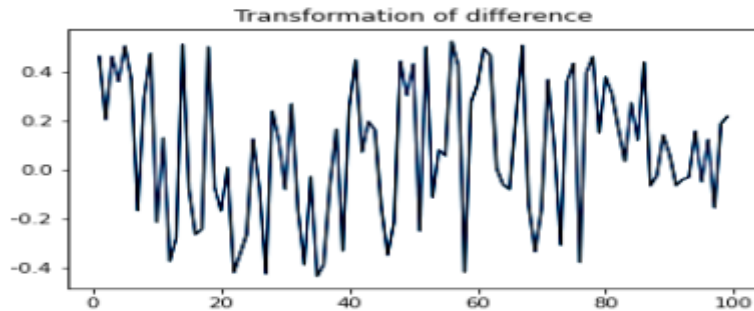
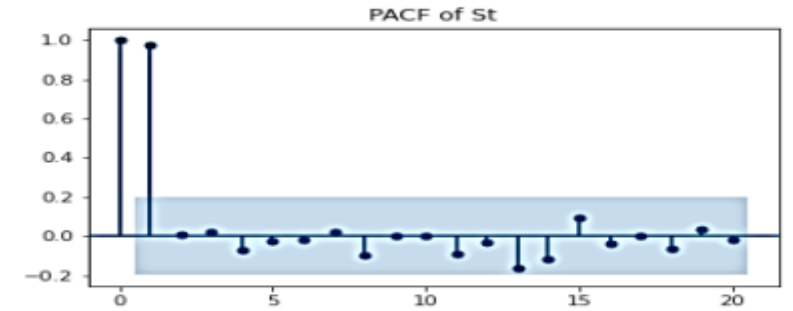
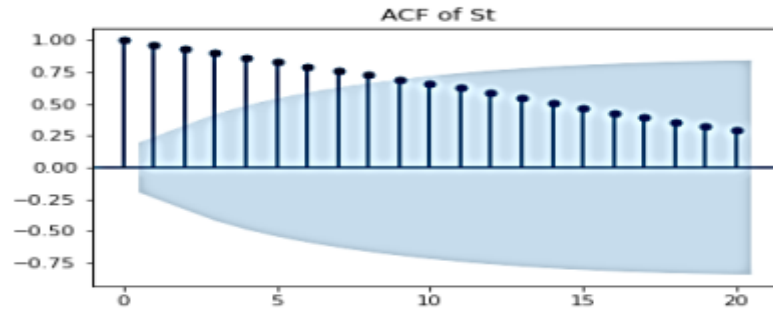
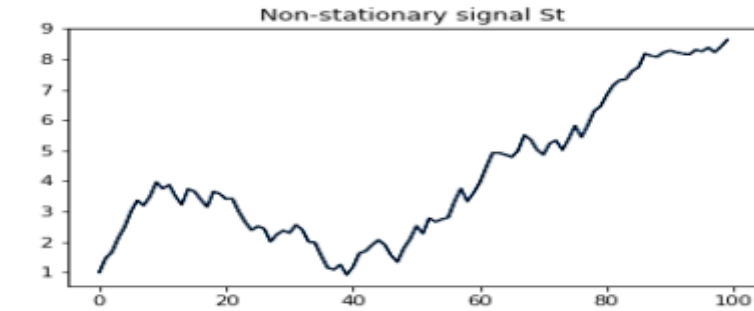
A slowly decaying ACF plot may indicate the presence of a trend in the data.

A sharp cutoff in the PACF plot at a particular lag may indicate the presence of a seasonal pattern in the data.

A slowly decaying PACF plot may indicate the presence of seasonality in the data.

A sharp cutoff in the ACF plot at a particular lag suggests that a time series model with a corresponding number of lags would be a good fit.

Example: Non-Stationary to Stationary and its impact



Stationarity

Definition

If $\{y_t\}$ is a **stationary time series**, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

A **stationary series** is:

- roughly horizontal
- constant variance
- no patterns predictable in the long-term

Definition

If $\{y_t\}$ is a stationary time series, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

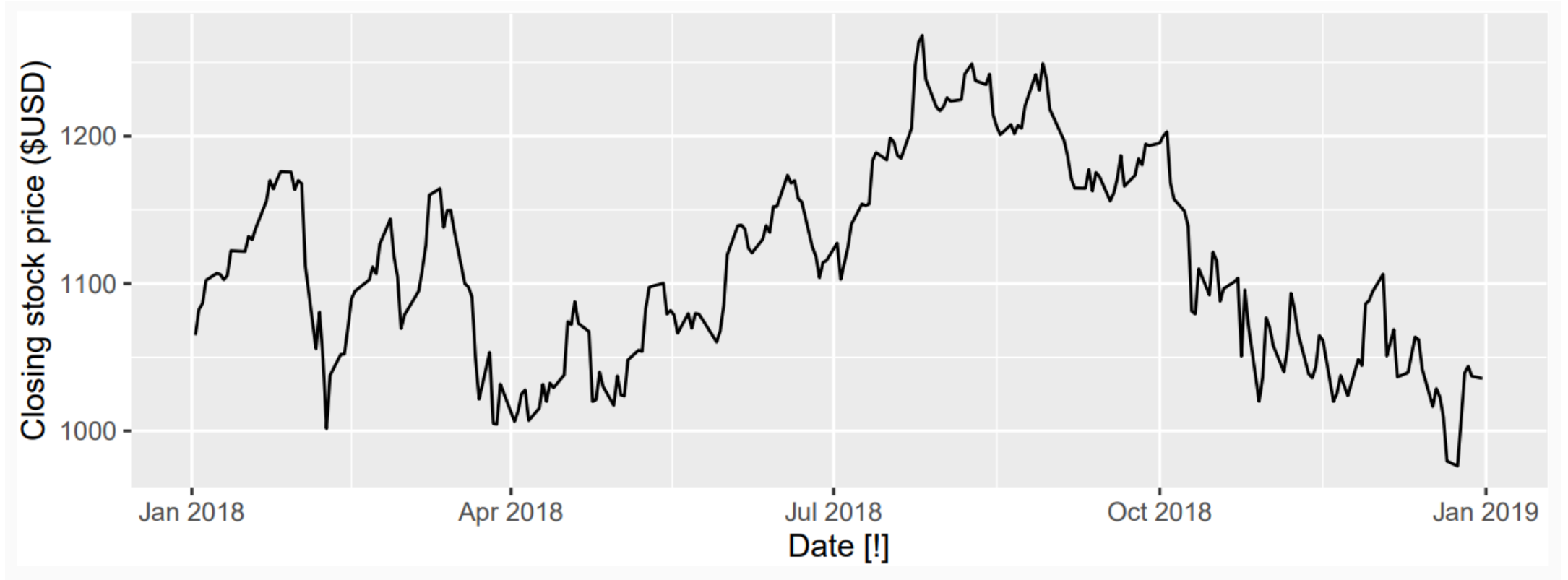
- Transformations help to stabilize the variance.
- For ARIMA modelling, we also need to stabilize the mean.

Non-stationarity in the mean

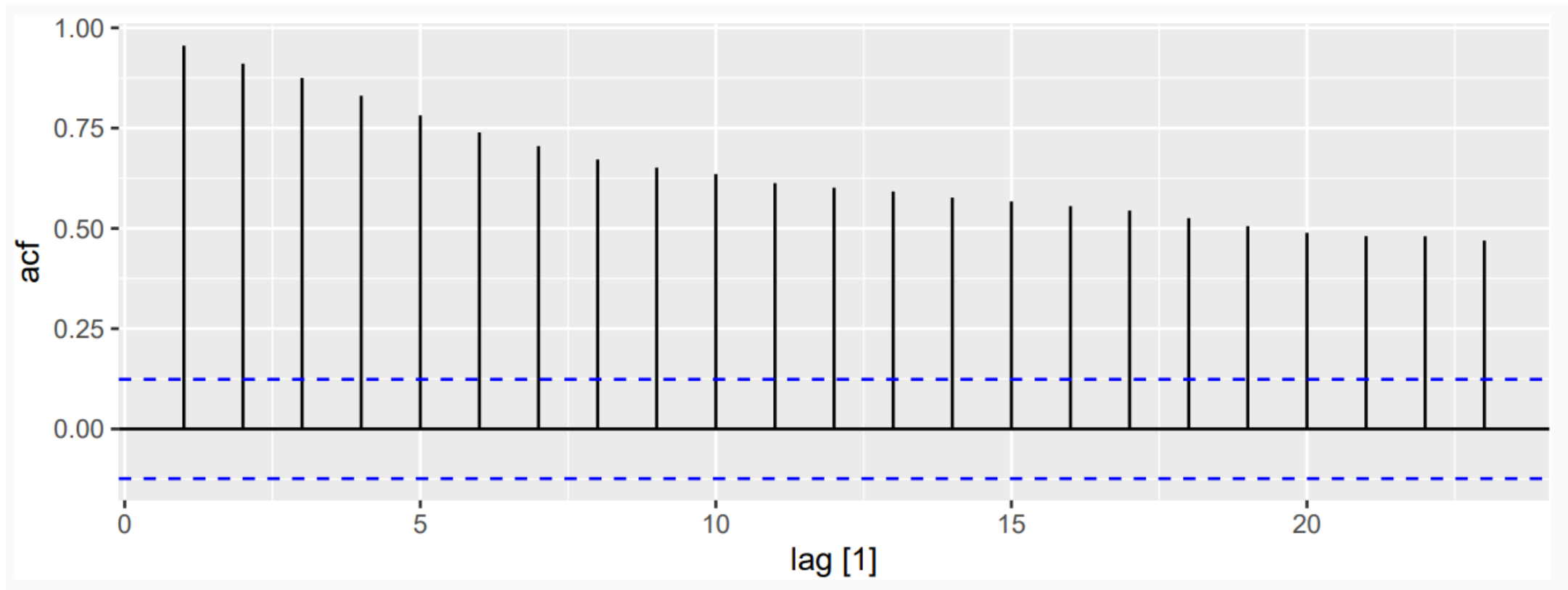
Identifying non-stationary series

- time plot.
- The ACF of stationary data drops to zero relatively quickly
- The ACF of non-stationary data decreases slowly.
- For non-stationary data, the value of r_1 is often large and positive.

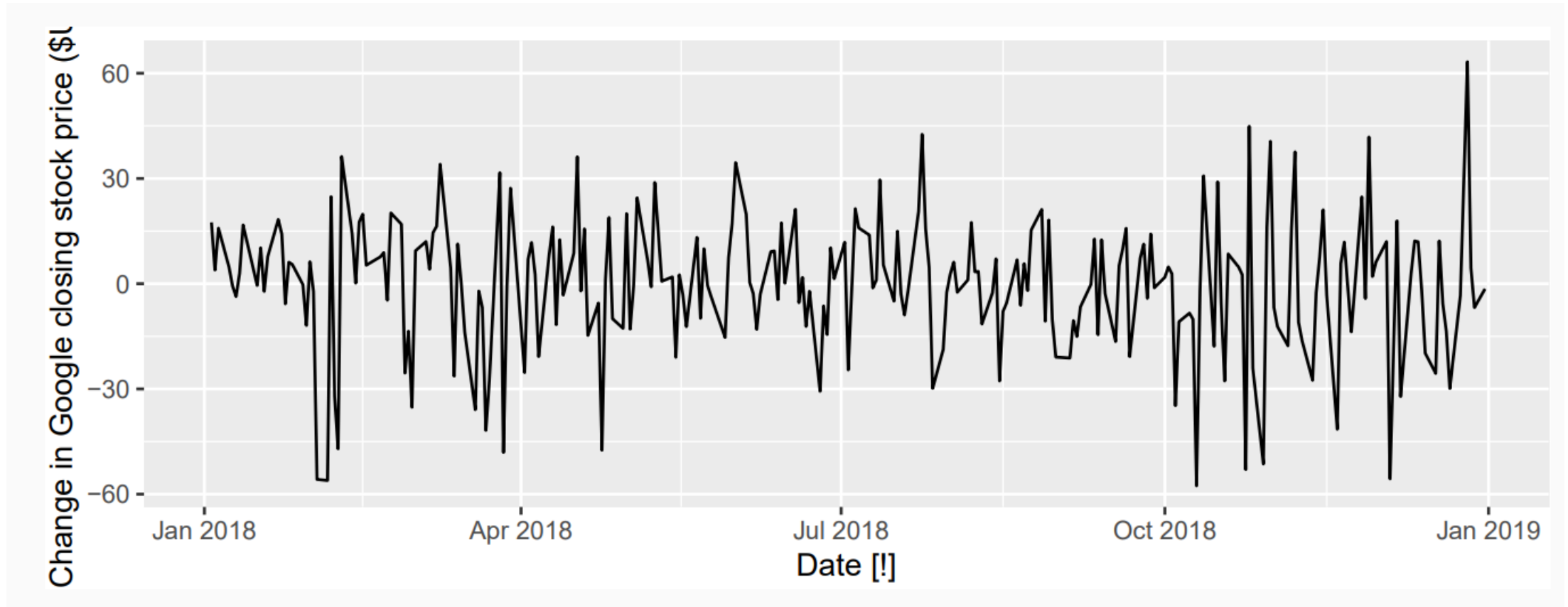
Example: Google stock price



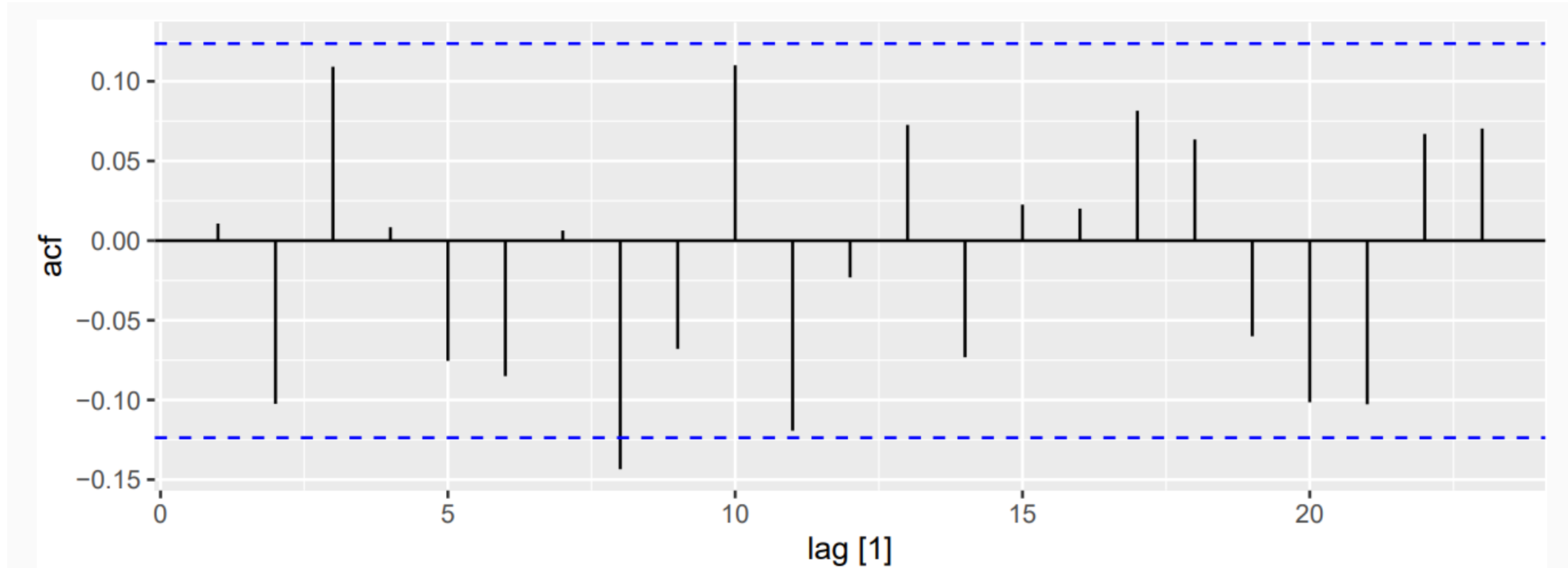
Example: Google stock price



Example: Google stock price



Example: Google stock price



Seasonal Index Example

Q.

Month	Demand			Average 2003-2005	Average Monthly	Seasonal Index
	2003	2004	2005			
Jan	80	85	105	90	94	90/94
Feb	70	85	85	80	94	80/94
Mar	80	93	82	85	94	
Apr	90	95	115	100	94	
May	113	125	131	123	94	
Jun	110	115	120	115	94	
Jul	100	102	113	105	94	
Aug	88	102	110	100	94	
Sept	85	90	95	90	94	
Oct	77	78	85	80	94	
Nov	75	72	83	80	94	
Dec	82	78	80	80	94	

Seasonal Index Example

Month	Demand			Average 2003-2005	Average Monthly	Seasonal Index
	2003	2004	2005			
Jan	80	85	105	90	94	0.957
Feb	100	102	115	105	94	
Aug	88	102	110	100	94	
Sept	85	90	95	90	94	
Oct	77	78	85	80	94	
Nov	75	72	83	80	94	
Dec	82	78	80	80	94	

Seasonal index =
$$\frac{\text{average 2003-2005 monthly demand}}{\text{average monthly demand}}$$

$$= 90/94 = .957$$



Seasonal Index Example

Month	Demand			Average	Average	Seasonal
	2003	2004	2005	2003-2005	Monthly	Index
Jan	80	85	105	90	94	0.957
Feb	70	85	85	80	94	0.851
Mar	80	93	82	85	94	0.904
Apr	90	95	115	100	94	1.064
May	113	125	131	123	94	1.309
Jun	110	115	120	115	94	1.223
Jul	100	102	113	105	94	1.117
Aug	88	102	110	100	94	1.064
Sept	85	90	95	90	94	0.957
Oct	77	78	85	80	94	0.851
Nov	75	72	83	80	94	0.851
Dec	82	78	80	80	94	0.851

Seasonal Index Example

Month	Demand 2003	Demand 2004	Demand 2005	Average 2003-2005	Average Monthly	Seasonal Index
Jan						0.957
Feb						0.851
Mar						0.904
Apr						1.064
May						1.309
Jun						1.223
Jul						1.117
Aug						1.064
Sept						0.957
Oct						0.851
Nov						0.851
Dec	82	78	80	80	94	0.851

Forecast for 2006

Expected annual demand = 1,200

Jan

$$\frac{1,200}{12} \times 0.957 = 96$$

Feb

$$\frac{1,200}{12} \times 0.851 = 85$$

	Demand						
Month	2003	2004	2005	Average 2003-2005	Average Monthly	Seasonal Index	Forecast for 2006
Jan	80	85	105	90	94	0.960285	96.02845
Feb	70	85	85	80	94	0.853586	85.35862
Mar	80	93	82	85	94	0.906935	90.69354
Apr	90	95	115	100	94	1.066983	106.6983
May	113	125	131	123	94	1.312389	131.2389
Jun	110	115	120	115	94	1.22703	122.703
Jul	100	102	113	105	94	1.120332	112.0332
Aug	88	102	110	100	94	1.066983	106.6983
Sep	85	90	95	90	94	0.960285	96.02845
Oct	77	78	85	80	94	0.853586	85.35862
Nov	75	72	83	77	94	0.81802	81.80202
Dec	82	78	80	80	94	0.853586	85.35862
!	1050						
Average	87.5	93.3	100				
Expected annual demand for 2006					1200		

Calculate the seasonal index of quarterly sales (\$'000s) for the data given below.

Year	Quarter 1	Quarter 2	Quarter 3	Quarter 4
2003	72	64	63	75
2004	75	66	64	89
2005	76	68	67	95

Aug

68.5

73.5

76.5

Aug

74.3

66

64.6

86.3

Index

74.3

$\frac{74.3}{72.8} = 1.02$

0.906

0.887

1.18

Overall Avg

$= \frac{68.5 + 73.5 + 76.5}{3}$

$= 72.8$

ACF Problem Calculation

Imagine we have the following returns for Microsoft prices:

Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
5%	1%	-2%	3%	-4%	6%	2%	-1%	-3%	4%

ACF Calculation

Let's suppose we want to compute the autocovariance at lag 1. You will need the returns up to day 10, and the 1-period lagged returns up to day 9.

Thus, you have the following data structure for returns on days 10 and 9:

Variable	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
r_t	5%	1%	-2%	3%	-4%	6%	2%	-1%	-3%	4%
r_{t-1}	.	5%	1%	-2%	3%	-4%	6%	2%	-1%	-3%

ACF Calculation

Do you get to see the difference between the 2 variables?

The second one is the first lag of X_t .

Now, since the 2 variables have different dimensions (the first one has 10 observations, while the second one has 9), we are going to use data from day 2 onwards.

Consequently, our data is as follows:

Variable	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
r_t	1%	-2%	3%	-4%	6%	2%	-1%	-3%	4%
r_{t-1}	5%	1%	-2%	3%	-4%	6%	2%	-1%	-3%

The covariance between these 2 variables will be the autocovariance of the returns at lag 1.

ACF Calculation

$$\sum_{t=}$$

Remember the autocovariance formula:

$$\gamma_s = \frac{\sum_{t=1}^T (r_t - \bar{r}) (r_{t-s} - \bar{r})}{N - 1}$$

- We can see that the average return is the same for both returns, in our case, for returns up to day 10 and up to day 9.
- **As we explained before, autocovariance and autocorrelation functions are applied only to stationary time series.**
- Consequently, not only the **variance but also the mean is a unique value for the whole span.** That's why the mean is the same for any lag of the price returns.
- The mean of the Microsoft price returns is 1.1%.

ACF Calculation

Variable	r_t	r_{t-1}	$(r_t - \bar{r})$	$(r_{t-1} - \bar{r})$	$(r_t - \bar{r})(r_{t-1} - \bar{r})$
Day 2	1%	5%	-0.100%	3.900%	-0.004%
Day 3	-2%	1%	-3.100%	-0.100%	0.003%
Day 4	3%	-2%	1.900%	-3.100%	-0.059%
Day 5	-4%	3%	-5.100%	1.900%	-0.097%
Day 6	6%	-4%	4.900%	-5.100%	-0.250%
Day 7	2%	6%	0.900%	4.900%	0.044%
Day 8	-1%	2%	-2.100%	0.900%	-0.019%
Day 9	-3%	-1%	-4.100%	-2.100%	0.086%
Day 10	4%	-3%	2.900%	-4.100%	-0.119%

ACF Calculation

The autocovariance is just the sum of the last column values divided by (N-1, equal to 8), which results in -0.046%.

$$\begin{aligned} & \text{VAR}(r_t) \\ &= \frac{\sum (r_t - \bar{r})^2}{(N-1)} \end{aligned}$$

Variable	r_t	r_{t-1}	$(r_t - \bar{r})$	$(r_{t-1} - \bar{r})$	$(r_t - \bar{r})(r_{t-1} - \bar{r})$
Day 2	1%	5%	-0.100%	3.900%	-0.004%
Day 3	-2%	1%	-3.100%	-0.100%	0.003%
Day 4	3%	-2%	1.900%	-3.100%	-0.059%
Day 5	-4%	3%	-5.100%	1.900%	-0.097%
Day 6	6%	-4%	4.900%	-5.100%	-0.250%
Day 7	2%	6%	0.900%	4.900%	0.044%
Day 8	-1%	2%	-2.100%	0.900%	-0.019%
Day 9	-3%	-1%	-4.100%	-2.100%	0.086%
Day 10	4%	-3%	2.900%	-4.100%	-0.119%

ACF Calculation

Calculation of the autocorrelation with an example

Let's follow the same exercise and compute the autocorrelation of the Microsoft price returns up to day 10 at lag 1.

The **autocorrelation is the autocovariance divided by the variance.**

The variance of **Microsoft price returns up to day 10 is 0.121%.**

$$\rho_s = \frac{\text{Cov}(r_t, r_{t-s})}{\text{Var}(r_t)} = \frac{-0.046}{0.121} = \frac{\sum (r_t - \bar{r})(r_{t-s} - \bar{r})}{\sum (r_t - \bar{r})^2}$$

$\rho_s = -0.38$

ACF Problem :2

Consider the following time series data. Compute the ACF for lag 1.

Day 1	Day 2	Day 3	Day 4
4	2	3	5

Partial Autocorrelation Function-PACF

- A **partial autocorrelation** is a summary of the relationship between an observation in a time series with observations at prior time steps with the relationships of **intervening observations removed**.
- The **autocorrelation** for an observation and an observation at a prior time step is comprised of both the direct correlation and indirect correlations.
- These **indirect correlations** are a linear function of the correlation of the observation, with observations at intervening time steps.
- It is these **indirect correlations** that the **partial autocorrelation function** seeks to remove.

